D07 · Supply Chain Reference Architecture v2.2

Warehousing, OMS & Fulfillment

The full-domain reference · mrsupplychain.ai

Warehousing, OMS & Fulfillment is the discipline where every promise the business made becomes a physical act. Planning decides what should exist and logistics decides what it costs to move; D07 decides whether the right unit, in the right quantity, with the right paperwork, actually leaves the building today. It is the only domain the customer experiences directly — and the one where LPPH, pick accuracy, and perfect order rate settle whether they are delighted or disappointed.

The interface between customer demand and physical execution.

HOW TO READ THIS DOMAIN

1 · Warehouse Operations & WMS. The physical design and systems that decide 80 LPPH or 50

2 · Order Management System (OMS/ATP/CTP). What is promised, from which node, and who wins when stock is scarce

3 · Pick-Pack-Ship & Fulfillment Execution. Turning a committed order into a packed, labeled, dispatched shipment

4 · Cross-Docking & Flow-Through. Eliminating storage where it adds no value

5 · Warehouse Automation & AMR/Robotics. From labor-intensive to capital-intensive — and when that is justified

STAGE 1 · FOUNDATION · 01 OF 06

Warehouse Operations & WMS

The operational foundation of the warehouse — the physical design, the systems, and the processes that determine whether the warehouse runs at 80 LPPH or 50 LPPH, at 99.9% inventory accuracy or 94%.

01 Warehouse layout design & slotting optimization

Layout design arranges receiving, storage, picking, packing and shipping zones, aisles and equipment to make flows short and uncongested; slotting assigns each SKU to the location that minimizes picking travel within its size, weight and compatibility constraints.

Warehouse layout: the architecture that determines operational efficiency

Warehouse layout design — the physical distribution of receiving, storage, picking, packing, and dispatch zones — is the single highest-impact decision on warehouse productivity. A poorly designed layout generates kilometers of unnecessary walking, congestion in high-activity zones, and picking times that could be reduced 20–40% with a reconfiguration.

Layout design principles

(1) High-velocity SKUs near the dispatch zone (reduces total picking travel). (2) Separate inbound and outbound flows to avoid congestion. (3) Aisle width calibrated to handling equipment (counterbalance vs. narrow-aisle vs. VNA). (4) Rack height optimized for storage density vs. access speed. (5) Value-added zones (labeling, kitting) positioned within the natural process flow.

Slotting: assigning each SKU to its optimal location

Slotting is the assignment of each SKU to the specific warehouse location that minimizes total picking travel — considering the SKU's picking frequency, size and weight, and compatibility restrictions (products that cannot be stored together). Re-slotting must be performed at least semi-annually — portfolio turnover means the optimal slotting of 12 months ago is no longer optimal today.

Intermediate vs. Advanced

Intermediate: applies defined slotting rules, proposes location changes when inefficiencies are detected.Advanced: designs the warehouse layout with correct flow and density principles; implements the optimized slotting process; leads the semi-annual re-slotting based on picking frequency analysis.

BEST PRACTICES

Frequency-based slotting delivers the fastest ROI of any warehouse initiative

In most distribution centers a small share of SKUs (often 10–20%) generates the majority of picking movements. Moving those SKUs to the shortest-travel zone has an immediate productivity impact.

Semi-annual re-slotting is the process that keeps the optimal slotting current

The SKU portfolio changes continuously — NPIs entering, discontinued products, seasonal shifts. The optimal slotting of 12 months ago is no longer optimal if the portfolio has changed significantly.

Warehouse layout must be designed before selecting equipment

The equipment type (counterbalance, narrow-aisle, VNA) determines the minimum aisle width — and aisle width determines available storage density. This sequence cannot be reversed without costly consequences.

Include compatibility restrictions in slotting — not just frequency

A slotting that only optimizes by picking frequency may place a food product and a chemical product together. Compatibility constraints (temperature, flammability, food vs. non-food) are hard constraints the slotting must respect.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> **Illustrative case: Warehouse re-slotting — distribution DC, 8,400 active SKUs, 280 picking operators** — Analysis reveals the top 200 highest-velocity SKUs are distributed throughout the warehouse with no frequency-based grouping.

Slotting action	SKUs affected	Productivity impact
Relocation of top 200 SKUs to golden zone (front, mid-level)	200	Picking travel for top SKUs: -62% · These SKUs represent 68% of all movements
Consolidation of slow-moving SKUs into deep storage zone	1,200	Space freed in mid zone: +18% for A-items · Space-neutral DIO impact
Bulky SKUs assigned to bottom level · Light SKUs to top level	3,400	Ergonomics improved · Handling incidents: -28%

Result: Picking productivity: 68 lines/hr -> 89 lines/hr (+31%). Walking distance/operator/day: 14.2 km -> 9.1 km (-36%). Re-slotting took 3 days to implement and generated $380K in annual picking cost savings.

WHAT YOU WOULD USE

Warehouse Planning

Manhattan Associates WMS — Slotting — Module: Slotting Optimization + Labor ManagementManhattan Associates' slotting module automatically calculates optimal locations for each SKU based on picking frequency, size, and compatibility restrictions.

SAP EWM — Warehouse Management — Module: Layout Design + Slotting RulesSAP Extended Warehouse Management with location management and configurable slotting rules by zone, SKU profile, and storage restrictions.

Analytics

Python (scipy / numpy) — Module: Slotting Optimization ModelPython-based slotting optimization models that calculate the optimal SKU-to-location assignment minimizing total picking travel distance.

HOW IT IS MEASURED

Lines Picked per Hour (LPPH) — picking productivity

Lines Picked per Hour (LPPH)Total picking lines completed / Total person-hours dedicated to picking

Benchmark: 60–100 LPPH for manual picking · 150–250 for assisted picking (voice, RF) · 200–400 for GTP An LPPH drop without a volume change indicates the slotting has deteriorated (high-velocity SKUs have migrated to suboptimal locations) or aisle congestion has increased.

Average Travel Distance per Pick

Average Travel Distance per PickTotal kilometers walked by picking operators / Total picks for the period

Benchmark: Continuous 5–15%/year reduction in active re-slotting programs · Typical baseline: 8–15 km/operator/day Travel distance per pick is the most direct slotting indicator. An increase without volume growth signals the optimal slotting is degrading as the active SKU portfolio changes.

Put the right SKU in the right place, and keep it there. Slot by velocity, respect compatibility, re-slot twice a year and use travel per pick as the alarm.

02 WMS implementation: receiving, put-away & inventory control

A Warehouse Management System (WMS) directs and records every physical movement — receiving against ASN, directed put-away, replenishment, picking, cycle counting — so that system inventory matches physical inventory by location, lot and serial.

The WMS: the nervous system of the warehouse

The Warehouse Management System (WMS) manages all physical movements in the warehouse — from truck receiving to order dispatch — with complete traceability of every inventory unit. Without a WMS, the warehouse operates on paper lists, operator tribal knowledge, and inventory accuracy systematically below 95%.

WMS core processes: receiving, put-away, and inventory

Receiving: verification of quantity and quality of received product against the supplier's ASN (Advanced Ship Notice). The WMS confirms receipt and updates available inventory in real time. Put-away: the WMS automatically assigns the storage location per slotting rules, optimizing space utilization and respecting compatibility constraints. Inventory control: the WMS maintains the exact location of every unit — lot, expiry date, serial number — and manages cycle counting to maintain inventory accuracy.

The WMS modules that generate the most value

(1) ASN-based receiving with scanner confirmation: eliminates receiving errors. (2) Directed put-away: the operator follows WMS instructions rather than deciding where to place the product. (3) Wave planning and batch picking: groups picks from multiple orders in a single warehouse pass. (4) Inventory accuracy management: automatically schedules cycle counting and manages discrepancies. (5) Labor management: measures each operator's productivity in real time.

Intermediate vs. Advanced

Intermediate: operates the WMS proficiently for assigned processes, reports detected discrepancies.Advanced: designs the WMS process with correct rule configuration; leads implementation; manages continuous improvement of WMS KPIs.

BEST PRACTICES

WMS-ERP integration is the prerequisite of WMS value

A WMS disconnected from the ERP generates two versions of inventory: the financial system's and the warehouse's. Real-time integration ensures that inventory available for sale (in the ERP/OMS) exactly reflects the physical warehouse inventory.

Configure directed put-away from day 1 — don't defer it

Undirected put-away (the operator decides where to place product) is the most frequent root cause of low IA and warehouse disorder. Directed put-away configuration is the most impactful process change of the WMS implementation.

Wave planning and batch picking are the highest-impact productivity modules

Single-order picking generates maximum walking distance. Batch picking — collecting multiple orders in a single warehouse pass — can reduce walking distance 40–60% with a well-configured WMS.

Warehouse team training is 40% of implementation success

A WMS poorly used by the team is worse than no WMS — because the system records incorrect transactions that degrade IA and generate fulfillment errors. Hands-on training for all operators before go-live is non-negotiable.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: WMS implementation — pharmaceutical distribution DC, paper-based operation — Current operation processes 2,400 lines/day with 85 operators. IA is 91.4% and urgent order processing time is 4.2 hours.

WMS module implemented	Timeline	Measured impact
ASN receiving + scanner confirmation (vs. paper)	Month 1–2	Receiving errors: 18/month -> 1/month · Receiving time per load: -38%
Directed put-away + FEFO rules (pharma)	Month 2–3	Product expiry from incorrect put-away: 0 incidents in 12 months post-go-live
Wave planning + batch picking for standard orders	Month 3–4	LPPH: 54 -> 78 lines/hr (+44%) · Operators required: 85 -> 68 (-20%)
Inventory accuracy management + daily cycle count	Month 4–5	IA: 91.4% -> 98.6% (+7.2 pp)

Result: 6-month total: LPPH +44%, IA +7.2 pp, urgent order time: 4.2 hr -> 1.8 hr. WMS ROI at 12 months: $1.4M in labor efficiency + $280K in error and expiry reduction.

WHAT YOU WOULD USE

WMS Platforms

SAP EWM (Extended Warehouse Management) — Module: WMS with Full SAP IntegrationThe reference WMS for SAP S/4HANA companies. Native integration with MM, SD, and PP. Supports multiple warehouse types and picking modalities.

Manhattan Associates WMS — Module: Best-in-Class WMS for Complex OperationsWorld-class WMS for high-complexity operations. Advanced labor management, slotting, and wave planning modules.

Blue Yonder Warehouse Management — Module: WMS + Labor Management + SlottingBlue Yonder WMS with advanced labor management and slotting optimization capabilities.

Mid-Market

Deposco / Extensiv (formerly 3PL Central) — Module: WMS for Mid-Market WarehousesWMS for mid-size warehouses with strong functionality-to-cost ratio.

HOW IT IS MEASURED

WMS Transaction Accuracy % (WMS transactions executed without error)

WMS Transaction Accuracy %(WMS transactions executed correctly without manual correction / Total transactions) × 100

Benchmark: >99.5% transaction accuracy in well-configured WMS processes A WMS accuracy <98% indicates transactions are being executed incorrectly with regularity — from master data problems, process errors, or omitted transactions.

Dock-to-Stock Time (time from truck arrival to product available in the system)

Dock-to-Stock TimeAverage time from receiving truck arrival to put-away confirmation and inventory availability in the WMS

Benchmark: <2 hours for standard receipts · <4 hours for high-complexity or quality-inspection receipts A long dock-to-stock time delays inventory availability for fulfillment — especially critical when the received product is needed to cover an urgent order.

Implement the WMS for accuracy first. Scan-confirmed receiving, directed put-away, cycle counting and real-time ERP integration make inventory trustworthy — everything else builds on that.

03 FEFO, lot traceability & quality management in the warehouse

FEFO (first expired, first out) rotates stock by expiry date; lot traceability links every unit to its source lot and destination; warehouse quality management blocks, inspects and releases lots under documented rules.

FEFO and lot traceability: the quality control that protects the end consumer

FEFO (First Expired, First Out) is the inventory rotation rule that ensures the product with the nearest expiry date leaves the warehouse first — before product with a later expiry. Without FEFO, the warehouse may dispatch soon-to-expire product while fresher product sits at the back of the shelf — resulting in expired products at the point of sale and potentially consumer risk.

Inventory rotation regimes

FIFO (First In, First Out): first in, first out — without considering expiry date. Correct for products without an expiry date. FEFO: first to expire, first out — correct for all products with expiry dates (food, pharmaceuticals, cosmetics). LIFO (Last In, First Out): only valid for certain accounting inventories — rarely used in the physical warehouse operation.

Lot traceability: tracking product from supplier to consumer

Lot traceability is the ability to track each product unit from its origin (supplier, production lot) to its final destination (customer, point of sale, consumer). Traceability is a regulatory requirement in pharmaceuticals (COFEPRIS, NOM-059-SSA1) and expected practice in food (NOM-251-SSA1 hygiene practices) — and the essential tool for managing a product recall quickly and precisely.

Intermediate vs. Advanced

Intermediate: applies FEFO rules in daily operations, verifies the WMS correctly manages lots and expiry dates.Advanced: designs the end-to-end traceability system with WMS-ERP integration; leads recall simulations; manages the lot blocking and disposition process for quality issues.

BEST PRACTICES

FEFO must be a hard WMS rule — not a suggestion

A WMS that “suggests” the FEFO location but allows the operator to select another location does not guarantee FEFO. The correct implementation locks out other locations until the FEFO location is empty.

Lot traceability requires capturing the lot at receiving — it cannot be added later

The lot and expiry date must be captured at receiving — when the product arrives with its supplier label. Attempting to add lot information after the product has been stored is impractical at scale.

Run recall simulations at least twice a year

A real recall is not the time to discover traceability gaps. Annual (or semi-annual) simulations verify traceability coverage and recall execution time before it's needed for real.

Integrate warehouse quality management with the company's QMS

A lot blocked for a quality issue detected in the warehouse must be automatically reflected in the QMS and ERP — so blocked inventory cannot be sold or dispatched while the investigation is managed.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: FEFO implementation with lot traceability — food distributor, 1,200 perishable SKUs — The distributor has 12 expired product complaints at POS per month. Current WMS does not manage FEFO.

Initiative	Implementation	Result
FEFO in the WMS: system only proposes locations with the nearest expiry date	Month 1–2	Expired product complaints at POS: 12/month -> 0 in 3 months post-implementation
Lot traceability: each receipt captures the supplier lot number and expiry date	Month 2–3	Traceability coverage: 0% -> 94% of receipts with lot captured
Automatic SLOB alert: WMS alerts when a product has <14 days of remaining shelf life in stock	Month 3	First month: 48 near-expiry alerts -> Action: promotions or supplier return

Result: Expired POS complaints: from 12/month to 0 in 3 months. Lot traceability enabled the first recall simulation in 4.2 hours (vs. a typical 8-hour internal target).

WHAT YOU WOULD USE

Traceability & Quality

SAP EWM + QM (Quality Management) — Module: FEFO + Lot Traceability + Quality InspectionNative SAP EWM integration with the Quality Management (QM) module manages FEFO, lot traceability, quality inspections at receiving, and lot blocking for quality issues.

TraceLink / Antares Vision (rfxcel) — Module: Pharmaceutical Serialization + TraceabilityTraceability and serialization platforms for the pharmaceutical industry. Comply with FDA (Drug Supply Chain Security Act) and COFEPRIS traceability requirements (NOM-059-SSA1).

HOW IT IS MEASURED

FEFO Compliance % (picks executed following the FEFO rule)

FEFO Compliance %(Picks that took product from the location with the nearest expiry date / Total picks of expiry-date products) × 100

Benchmark: >99% FEFO compliance for pharmaceutical products · >97% for food products FEFO compliance <95% indicates the WMS is not forcing selection of the correct location — or operators are ignoring WMS instructions. Both problems generate expired products at POS.

Lot Traceability Coverage % (% of inventory with full lot traceability)

Lot Traceability Coverage %(Inventory units with lot number and expiry date captured / Total inventory units of expiry-date products) × 100

Benchmark: 100% for pharmaceuticals (regulatory requirement) · >95% for food Complete traceability is the prerequisite of an effective recall. Without lot traceability, a recall requires withdrawing all inventory of a product from all channels — rather than just the affected lots.

Enforce rotation and capture traceability at the source. Lot and expiry at receiving, FEFO as a hard rule, quality holds the system respects, and recalls rehearsed before they are real.

04 Labor management & workforce planning in the warehouse

Warehouse labor management plans staffing from forecast volume and engineered standards, measures productivity against those standards, and develops the team through feedback, incentives and training.

Warehouse labor management: the largest variable cost in the operation

Labor represents 60–70% of the operating cost of a non-automated warehouse. Effective warehouse labor management — staffing planning, per-operator productivity measurement, absenteeism management, and training — has a direct impact on operating cost and service level.

Components of warehouse labor management

(1) Engineered labor standards: the standard time for each warehouse activity (picking a line, receiving a pallet, pallet picking, etc.), calculated by time study or WMS data. (2) Labor management system (LMS): the system that measures each operator's productivity vs. standard and generates the performance report. (3) Staffing model: the headcount required per shift based on planned volume and productivity standards. (4) Incentive program: the incentive system that rewards high productivity and work quality.

The staffing model: how many operators per shift

Correct warehouse staffing: (1) Projected volume for the day/week (picking lines, receiving pallets). (2) Standard productivity per operator (LPPH). (3) Utilization factor (% of productive vs. total time — typically 75–85%). Operators required = Volume / (LPPH × Utilization factor × Shift hours). Understaffing generates costly overtime; overstaffing generates paid idle time.

Intermediate vs. Advanced

Intermediate: manages the team's productivity, reports labor performance KPIs.Advanced: designs the staffing model with correct productivity standards; implements the LMS; manages the incentive and team development programs.

BEST PRACTICES

Implement individualized productivity feedback — not just the team average

An operator who sees their daily productivity vs. standard has information to improve. A team that only sees the DC average has no individual reference to act on.

Volume-based staffing model is better than fixed staffing

A DC that adjusts headcount to projected daily volume (more operators on high-volume days, fewer on low-volume days) has a structurally lower labor cost than one with fixed staffing for the worst case.

The productivity incentive program has the best ROI of any warehouse management initiative

A well-designed incentive program (bonus for LPPH above standard, top performer recognition) improves average team productivity 10–20% with an incentive cost typically <3% of labor cost.

Talent retention in the warehouse is as important as productivity

Warehouse turnover rate impacts productivity because new operators are at 60–80% of standard during the first 3 months. A warehouse with 40% annual turnover always has 20–25% of its team on a learning curve.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> **Illustrative case: Labor Management System implementation — DC with 120 picking operators** — The DC has no individual productivity visibility. LPPH variability between operators is enormous.

Pre-LMS metrics	Value	Post-LMS action and result
LPPH range between operators (best vs. worst)	HIGH	42–98 LPPH (133% variation) · No individual feedback
High-productivity operators identified and recognized	HIGH	0% — all treated equally · No incentive system

Pre-LMS metrics	Value	Post-LMS action and result
Productive vs. presence time (utilization)	MEDIUM	68% average utilization (32% unmanaged idle time)
Productivity-based incentive program	HIGH	No program · High turnover: 38%/year

Result: Post-LMS (12 months): Average LPPH: 64 -> 82 (+28%). Utilization: 68% -> 79%. Turnover: 38% -> 24%. Daily individualized feedback and the incentive program (weekly bonus for operators above standard) generated the greatest impact. Operators with consistently low LPPH received targeted training.

WHAT YOU WOULD USE

Labor Management

Manhattan Associates WMS — Labor Management — Module: Engineered Standards + LMS + Incentive ManagementManhattan Associates labor management module with per-task productivity standards, real-time individual measurement, and incentive program management.

SAP EWM — Labor Management — Module: WMS-Integrated Labor TrackingSAP EWM labor management module integrated with the WMS. Measures productivity per operator, per task, and per shift.

HOW IT IS MEASURED

Labor Utilization % (% of operator presence time that is productive)

Labor Utilization %(Productive operator hours / Total paid presence hours) × 100

Benchmark: 75–85% utilization in well-managed manual picking operations Utilization <70% indicates significant idle time (waits, unassigned operators, flow problems) being paid without generating value.

LPPH vs. Engineered Standard % (actual productivity vs. calculated standard)

LPPH vs. Engineered Standard %(Actual LPPH of operator or shift / Calculated standard LPPH) × 100

Benchmark: >95% of standard for experienced operators · 80–90% during the learning curve (first 3 months) Comparing actual LPPH vs. engineered standard identifies operators below standard who need additional training, and those above who deserve recognition.

Plan, measure and keep your people. Engineered standards, volume-based staffing, daily individual feedback and a retention strategy are the cheapest productivity program a warehouse has.

05 Value-added services (VAS) & postponement in the warehouse

Value-added services (VAS) are customization steps performed in the warehouse — labeling, kitting, repackaging, serialization, personalization; postponement delays final product differentiation until the last responsible point in the chain.

Value-added services: transforming the warehouse into a customization center

Value-Added Services (VAS) are the customization or adaptation activities performed in the warehouse before dispatch — labeling, kitting, promotional kit assembly, repackaging, serialization, and personalization. VAS converts the warehouse from a storage-and-movement node into a value-creation node — with a direct impact on supply chain flexibility and net margin.

Most common VAS types in logistics operations

(1) Labeling and re-labeling: retail price label, destination label, pharmaceutical serialization. (2) Kitting: assembly of multiple SKUs into a kit or bundle for a specific order or promotion. (3) Repackaging: changing product packaging from warehouse format to point-of-sale format. (4) Custom pick & pack: preparing orders with specific configurations for each customer. (5) In-transit quality control: inspection of received merchandise before storage or dispatch.

Postponement: deferring product differentiation as late as possible in the chain

Postponement is the strategy of keeping product in its most generic form (no language label, no regional configuration) as long as possible in the chain — and differentiating it (labeling in the correct language, assembling the regional configuration) at the last node before the customer. Postponement reduces total inventory required (the generic product serves all markets) while maintaining the customization customers expect.

Intermediate vs. Advanced

Intermediate: executes assigned VAS with defined quality and productivity standards, reports VAS operation KPIs.Advanced: designs the VAS strategy with correct cost-benefit analysis; evaluates postponement opportunities in the portfolio; manages VAS integration into the WMS flow.

BEST PRACTICES

Postponement can reduce total inventory 20–40% in chains with high SKU diversity

A product with 4 regional variants (label in 4 languages) can be managed as 1 generic SKU in the central warehouse and differentiated at the regional DC or POS. Required inventory is significantly lower than 4 independent SKUs through demand aggregation.

Integrate VAS into the WMS flow from design — don't add it as a parallel process

VAS "outside the WMS" (operator assembles the kit without the WMS knowing) generates inventory discrepancies, lack of traceability, and difficulty measuring productivity and quality.

The make vs. buy analysis for VAS must include flexibility and response speed

A VAS performed in our warehouse can be executed with 24 hours of notice. The same VAS performed by the supplier requires 4 weeks. Response speed is a benefit of internal VAS that a purely financial cost analysis may underestimate.

Define VAS quality standards with the same criteria as finished product

A kit incorrectly assembled in the warehouse is a product quality defect — not a warehouse error. VAS quality standards must be as rigorous as the manufacturing process standards.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Kitting implementation in warehouse — electronics company, 12 promotional kits — The company launches 12 new promotional kits that must be assembled in the DC (1 tablet + 1 keyboard + 1 case) before dispatch.

Kitting process element	Design	KPI achieved
Dedicated kitting station with WMS integration	3 stations · WMS guides assembly and scanner-verifies each component	Kitting error rate: 0.08% (1 error in 1,250 kits) vs. 2.4% without scanner
Kitting productivity	Standard: 28 kits/hr per operator	Achieved: 31 kits/hr (111% of standard in month 2)
Kit traceability (each component's lot)	WMS captures each component's serial number in the kit	Full traceability: 100% of kits with identified components
Kit component inventory management	WMS inventory reservation for active kit components	0 kitting process stoppages due to missing components in months 2 and 3

Result: Kitting implemented in 4 weeks. Error rate: 0.08% vs. 2.4% without WMS guidance. Productivity: 31 kits/hr. Integrating kitting into the WMS was the most important decision — manual kitting outside the WMS generated component errors and traceability loss.

WHAT YOU WOULD USE

VAS & Postponement

SAP EWM — Warehouse Task Management — Module: VAS Integration + Kitting + LabelingSAP EWM manages VAS processes (kitting, labeling, repackaging) within the WMS flow, with full traceability and component inventory management.

Manhattan Associates WMS — Value-Added Services — Module: VAS Workflows + Quality ControlManhattan Associates VAS module with configurable workflows for each VAS type and scanner confirmation at each process step.

HOW IT IS MEASURED

VAS Quality Rate % (VAS executed without error or rework)

VAS Quality Rate %(VAS units executed correctly without rework / Total VAS units processed) × 100

Benchmark: >99.5% for kitting · >99.9% for labeling and serialization A VAS quality rate <98% indicates a VAS process with insufficient quality control — generating errors that reach the customer and are more costly to correct after the fact than to prevent at the source.

VAS Cost per Unit (cost of each VAS unit executed)

VAS Cost per UnitTotal VAS operation cost (labor + materials + overhead) / Total VAS units processed

Benchmark: Benchmark vs. the cost of the same VAS performed by the supplier (make vs. buy) · Declining trend indicates process efficiency improvement In-warehouse VAS only makes financial sense if the cost is lower than performing it at the supplier, plant, or by the customer — and if the flexibility it generates has strategic value.

Differentiate late, control tightly. Postpone where variants share a generic base, integrate VAS into the WMS, and hold it to manufacturing-grade quality standards.

Order Management System (OMS/ATP/CTP)

The decision layer that determines how each order is fulfilled — from which node serves it, to what is promised to the customer, to how conflicts are resolved when inventory is scarce.

01 OMS architecture: order capture, routing & orchestration

An Order Management System (OMS) captures orders from every channel, checks availability, decides which node fulfills each order — DC, store or supplier — and orchestrates execution with the WMS and TMS while keeping the customer informed.

The OMS: the brain that orchestrates fulfillment of every order

The Order Management System (OMS) manages the complete lifecycle of an order — from capture (when the customer places the order) to delivery confirmation. The OMS orchestrates which logistics network node will serve the order (DC, store, supplier dropship), verifies availability with ATP, generates the delivery promise, and coordinates execution with the WMS and TMS.

OMS core functions

(1) Order capture: receives orders from all channels (e-commerce, EDI, phone, app, physical store) in a single system. (2) ATP/Available-to-Promise: verifies inventory availability to fulfill the order. (3) Order routing: decides which network node will serve the order — optimizing cost, lead time, and inventory availability. (4) Order orchestration: coordinates order execution with the WMS (for dispatch) and TMS (for transport). (5) Order visibility: provides the customer and CS team real-time order status.

OMS vs. ERP vs. WMS: system boundaries

The ERP manages accounting inventory and the financial order cycle (invoice, accounts receivable). The WMS manages physical inventory movements in the warehouse. The OMS manages order business logic — what was promised, from where it's served, and what status it's in. All three layers must be integrated but have distinct responsibilities.

Intermediate vs. Advanced

Intermediate: processes orders in the OMS per defined routing rules, manages exceptions for non-routable orders.Advanced: designs OMS architecture with correct routing rules and ATP policies; leads implementation; manages integration with channel systems and execution systems.

BEST PRACTICES

The OMS is the most important system in the omnichannel architecture — not the WMS or TMS

The WMS and TMS are execution systems. The OMS is the decision system that defines how each order is served. Without an OMS that orchestrates correctly, the most sophisticated execution systems cannot deliver the omnichannel experience customers expect.

Real-time OMS-WMS-TMS integration is the prerequisite of promise accuracy

An OMS that makes a delivery promise without real-time confirmation from the WMS (is inventory available?) and the TMS (how long will transit take?) will generate promises that cannot be kept.

Define OMS routing rules with the commercial team — not just logistics

OMS routing rules have commercial implications (prioritize strategic customer) and logistical implications (nearest node, lowest cost). Rule design requires input from both teams.

Measure order routing accuracy — the % of orders routed to the optimal node

It's the metric that verifies whether the OMS routing engine is functioning correctly and whether the business rules reflect the company's current priorities.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> **Illustrative case: Omnichannel OMS implementation — retailer with 5 channels and 4 inventory nodes** — Retailer has 5 sales channels (web, app, store, call center, marketplace) and 4 nodes (national DC, regional DC, online stores, dropship). Without an OMS, each channel has its own routing logic and inventory management.

OMS functionality	Before (no OMS)	After (with OMS)
Unified inventory visibility across 4 nodes	HIGH	Siloed inventory by channel · Frequent overselling when same item sold in 2 channels
Automatic routing to lowest-cost/lead-time node	HIGH	Manual, ad-hoc rules · 28% of orders suboptimally routed
Unified delivery promise across all channels	MEDIUM	Generic lead time by channel · Not differentiated by node or destination
Real-time order status visibility for customer and CS	HIGH	0% real-time visibility · CS managed updates by email

Result: Post-OMS: Overselling eliminated. Optimal routing: 94% of orders to the correct node. Promise accuracy: 91.4% -> 97.2%. CS query resolution time: 8.4 min -> 2.1 min with real-time order status visibility.

WHAT YOU WOULD USE

OMS Platforms

Manhattan Active Omni — Module: Omnichannel OMS + ATP + Order RoutingThe reference OMS for complex omnichannel operations. Manages the complete order lifecycle from capture to delivery with intelligent routing based on cost, lead time, and inventory availability.

SAP S/4HANA — Sales Order Management + aATP — Module: ERP-Integrated OMS + Advanced ATPSAP S/4HANA with the aATP (Advanced Available-to-Promise) module as an OMS integrated with the ERP for companies that don't require a standalone OMS solution.

Salesforce Order Management — Module: Cloud OMS for B2C RetailSalesforce OMS for B2C retailers with native integration with Salesforce Commerce Cloud and Service Cloud.

HOW IT IS MEASURED

Order Routing Accuracy % (orders routed to the optimal node)

Order Routing Accuracy %(Orders routed to the lowest-cost + lead-time node per OMS policy / Total orders routed) × 100

Benchmark: >95% of orders routed to the optimal node by the OMS engine Incorrect routing creates two simultaneous problems: the order is served from a suboptimal node (higher cost or longer lead time) and the inventory at the optimal node remains unused.

Order Cycle Time (time from order capture to delivery)

Order Cycle TimeConfirmed delivery date/time - Order capture date/time

Benchmark: <24 hours for same-day e-commerce · <48 hours for next-day · <5 business days for standard B2B · Varies by channel promise Order cycle time is the KPI that measures the total speed of the end-to-end logistics chain. Differentiation by channel (same-day, next-day, standard) reflects the different value propositions to the customer.

Make the OMS the single brain of the order. One capture, one inventory view, routing rules written with the business, and real-time integration with execution.

02 ATP & CTP: real-time availability & promise management

Available-to-Promise (ATP) calculates what can be committed to a new order from stock and confirmed supply; Capable-to-Promise (CTP) extends the check to whether production or replenishment capacity can make it in time.

ATP and CTP: how much can I promise and when can I deliver it

ATP (Available-to-Promise) calculates in real time how much inventory is available to commit to new orders, considering current inventory, scheduled production or replenishment orders, and existing delivery commitments. CTP (Capable-to-Promise) goes further: if inventory is not available, CTP evaluates whether it can be produced in time to fulfill the order — integrating the capacity plan with promise management.

ATP vs. CTP: when to use each

ATP: inventory already exists (in the warehouse or in already-scheduled production) and the system calculates how much is available for new commitments. Applicable in all environments. CTP: inventory doesn't exist but can be produced. Requires integration with the production capacity plan. Applicable in make-to-order or configure-to-order environments where product is

manufactured specifically for the order.

Multi-echelon ATP: inventory visibility across all network nodes

An omnichannel ATP considers inventory across multiple nodes simultaneously: DCs, stores, in-transit, in production. An ATP that only sees DC inventory may promise inventory already committed from a store, or may not see in-transit inventory arriving tomorrow.

Intermediate vs. Advanced

Intermediate: consults and uses ATP in the customer delivery promise process, escalates situations where ATP cannot cover the order.Advanced: designs the ATP model with correct supply sources; implements CTP for make-to-order categories; manages the inventory allocation process under constraint.

BEST PRACTICES

The ATP is only as good as the data feeding it — data quality is the prerequisite

An ATP with inventory data updated 6 hours ago (by overnight batch process) cannot commit the inventory from the last 6 hours of transactions. Real-time ATP requires real-time data feeds.

Multi-node ATP requires all inventory sources to be integrated

An ATP that doesn't see in-transit, store, or production inventory is a partial ATP that underestimates real availability and generates order rejections or unnecessarily long lead time promises.

ATP cannot commit the same inventory to two channels simultaneously

The golden rule of ATP: one inventory unit can be committed to only one order at a time. An ATP that allows "overselling" (committing more inventory than available) generates delivery failures that are extremely costly to manage.

CTP requires integration of ATP with the production capacity plan

For CTP to work, the OMS must be able to query available production capacity and production lead time for each SKU in real time — requiring integration between the OMS and the APS (Advanced Planning System) or the ERP's MRP.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Real-time ATP implementation — electronics manufacturer, 3 inventory nodes — Manufacturer has inventory in 3 nodes (plant, national DC, in-transit) but ATP only queries the national DC.

Supply source	Current ATP status	Gap impact
National DC inventory	OK	Visible in current ATP · System baseline
Plant inventory (finished goods ready to ship)	HIGH	Not visible in ATP · Orders rejected for "out of stock" when 840 units available at plant
In-transit inventory (trucks with known ETA)	HIGH	Not visible in ATP · $2.4M of in-transit product not considered for delivery commitments
Scheduled production orders (in process)	MEDIUM	Partially visible · Only current-day production

Result: Post multi-node ATP implementation: Customer fill rate improved from 88.4% to 94.2% with no inventory changes — the inventory existed but the ATP couldn't see it. 840 units of "ghost inventory" at plant released for sales. $2.4M of in-transit inventory included in ATP logic for future delivery promises.

WHAT YOU WOULD USE

ATP & CTP

SAP S/4HANA — aATP (Advanced ATP) — Module: Multi-Node ATP + CTP + AllocationSAP S/4HANA's advanced ATP module queries inventory across multiple sources (warehouses, plants, in-transit, scheduled production) and executes CTP integrating the capacity plan.

Manhattan Active Omni — Real-Time ATP — Module: Omnichannel Inventory AvailabilityManhattan Associates real-time ATP system for omnichannel environments with multiple inventory nodes and multiple sales channels.

HOW IT IS MEASURED

ATP Accuracy % (delivery promises generated by ATP that are fulfilled)

ATP Accuracy %(Orders with fulfilled ATP promise / Total orders with ATP promise generated) × 100

Benchmark: >97% of ATP promises fulfilled · ATP accuracy <90% indicates incorrect assumptions in the ATP calculation Low ATP accuracy is the clearest indicator that the ATP system has incorrect assumptions — inventory the ATP considers "available" isn't in practice, due to unrecorded reservations or quarantine inventory not reflected.

ATP Response Time (time for the system to calculate ATP for an order)

ATP Response TimeTime from ATP request to system response (milliseconds or seconds)

Benchmark: <2 seconds for real-time e-commerce ATP · <30 seconds for complex multi-node ATP An ATP that takes >10 seconds to respond cannot be used in an e-commerce environment where the customer expects near-instant availability confirmation.

Promise from complete, current data. Include every legitimate supply source, remove reservations and holds, add CTP for make-to-order, and measure how often promises are kept.

03 Order prioritization, allocation & backorder management

Order prioritization and allocation decide which orders receive inventory when supply is short, using explicit rules — customer segment, contract, urgency, penalties — and manage backorders with reliable availability dates.

Order prioritization: deciding who receives inventory when there isn't enough for everyone

When available inventory is insufficient to fulfill all orders, the OMS must apply prioritization logic that assigns available inventory to orders in the correct sequence — based on customer strategic value, order urgency, channel, or SLA agreement. Without explicit prioritization logic, inventory is assigned first-come first-served (FIFO) — which rarely reflects the company's business priorities.

Order prioritization criteria

Most common prioritization criteria under inventory constraint: (1) Strategic customer (A-B-C customer segmentation): A customers always fulfilled first. (2) Channel: contractually committed channel takes priority over spot. (3) Order urgency: a hospital emergency order takes priority over a standard replenishment order. (4) Contractual penalty: orders whose non-fulfillment generates a higher penalty are prioritized.

Backorder management: managing the order when there's no stock

A backorder is an order that cannot be served immediately due to inventory shortage but the customer confirms they will wait. The OMS must manage the backorder: notify the customer of the projected availability date, reserve arriving inventory, and automatically fulfill the order when inventory becomes available.

Intermediate vs. Advanced

Intermediate: manages backorder orders per defined prioritization rules, communicates backorder status to the customer.Advanced: designs OMS prioritization and allocation rules; implements automatic backorder management; leads inventory triage in scarcity situations.

BEST PRACTICES

Define OMS prioritization rules with commercial and SC teams before scarcity occurs

Prioritization rules under scarcity are business decisions — not operational decisions. They must be approved in the S&OP as company policy before a real scarcity situation requires applying them under pressure.

Communicate backorders proactively to the customer — don't wait for the customer to ask

Proactive backorder communication (with projected availability date and reason) converts bad news into a customer service gesture. Reactive communication converts a delay into a complaint.

Backorder management requires the OMS to be integrated with the supply plan

The projected availability date for the backorder must be calculated with real production or replenishment plan data — not a generic “2–3 week” estimate. An OMS integrated with the APS can calculate the real availability date.

Analyze the backorder root cause — it’s a symptom of an upstream problem

A high backorder rate is a symptom of: insufficient safety stock, incorrect forecast, or supply constraint. The corrective action is in the inventory policy or the supply plan — not in the OMS.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Backorder management in scarcity — pharmaceutical manufacturer, 3 weeks without production due to equipment failure — Manufacturer has 180 active orders and only 40% of the inventory needed to cover them.

Prioritization rule applied	Orders affected	OMS decision
Priority 1: Hospitals and pharmacies with contractual SLA (penalty if not fulfilled)	42 orders · 100% of available inventory assigned first to this segment	42 orders fulfilled 100% · $0 contractual penalty
Priority 2: A-distributors (>80% of revenue)	38 orders · Remaining inventory assigned by order size	28 orders covered 100% · 10 orders in partial backorder (60–80%)
Priority 3: B and C distributors	100 orders · No inventory available	100 orders on backorder · Projected availability date: 18 days

Result: With OMS prioritization: $0 in contractual penalties. The 28 key A-distributors received available inventory. B and C distributors received proactive communication of the projected availability date 3 weeks in advance — minimizing damage to commercial relationships.

WHAT YOU WOULD USE

Backorder Management

SAP S/4HANA — Backorder Processing (BOP) — Module: Automated Backorder Management + Priority RulesSAP S/4HANA BOP module automatically manages backorders with configured prioritization rules, reallocates available inventory by priority, and updates projected availability dates.

Manhattan Active Omni — Order Promising — Module: Backorder + Promise ManagementManhattan system with backorder and delivery promise management integrating current inventory, in-transit, and the production plan.

HOW IT IS MEASURED

Backorder Rate % (orders that cannot be served immediately)

Backorder Rate %(Orders entering backorder / Total orders for the period) × 100

Benchmark: <3% for operations with well-managed inventory · <8% in planned demand peaks A sustained backorder rate >10% indicates inventory is not aligned with demand — from insufficient safety stock, incorrect forecast, or a structural supply constraint.

Backorder Fill Rate % (backorders fulfilled on the projected date)

Backorder Fill Rate %(Backorders fulfilled on the projected availability date communicated to the customer / Total backorders for the period) × 100

Benchmark: >90% of backorders fulfilled on the projected date · A backorder date that also isn’t met is a double failure The credibility of backorder management depends on the projected date communicated to the customer being accurate. A backorder whose date is updated 3 times generates more dissatisfaction than the original backorder.

Decide priorities before the shortage. Segment, approve the rules in S&OP, encode them in the OMS and give backordered customers dates they can trust.

04 B2B order management: EDI, vendor portals & contract compliance

B2B order management handles contract-based orders exchanged electronically — EDI purchase orders, acknowledgments, advance ship notices and invoices — and the customer-specific requirements and service-level agreements attached to them.

B2B order management: the complexity of business-to-business orders

B2B orders have very different characteristics from B2C orders: larger volume, subject to contracts with specific prices and conditions, transmitted via EDI (Electronic Data Interchange) in structured format, and compliance requirements with customer SLAs that include penalties for non-compliance. The B2B OMS must manage all this complexity in an automated way.

EDI in B2B order management

EDI is the standard for electronic exchange of business documents between companies — purchase orders (EDI 850/ORDERS), order confirmations (EDI 855), dispatch notices (EDI 856/DESADV), and invoices (EDI 810/INVOIC). Large retailers typically send orders via EDI and expects the supplier to respond with the same standards — including compliance with labeling requirements, delivery windows, and exact quantities.

B2B customer SLA compliance: OTIF and penalties

Major B2B customers — especially retailers — have very strict OTIF SLAs (often in the 95–98% range) with non-compliance penalties calculated as a percentage of the non-fulfilled value. The B2B OMS must manage these compliance rules and generate alerts when an order is at risk of non-compliance before it occurs.

Intermediate vs. Advanced

Intermediate: processes received EDI orders, manages exceptions when there are discrepancies between the order and available stock.Advanced: designs the EDI architecture with correct mappings; manages compliance with the most important B2B customer SLAs; implements OTIF risk alerts.

BEST PRACTICES

Well-configured EDI eliminates manual work in B2B order processing

A fully automated EDI flow (receipt -> mapping -> OMS -> WMS without manual intervention) frees the CS team to manage exceptions and relationships, not to process orders.

Configure OTIF risk alerts in the OMS for penalty customers

An order that has been in the WMS for 36 hours without having started picking for a customer with a 48-hour delivery window is an OTIF risk that can be managed if detected in time.

The 856 (ASN) sent to the retailer before dispatch is as important as the dispatch itself

Major retailers confirm ASN receipt before scheduling the dock appointment. An ASN sent late or with incorrect data can result in a rejection at the retailer's door — which counts as an OTIF non-compliance.

Integrate B2B customer SLA compliance into the S&OP process

OTIF of key B2B customers must be part of the S&OP dashboard — with visibility of at-risk penalties and root causes of non-compliances to take systemic corrective actions.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: EDI order management — FMCG supplier, 3 B2B retailers with strict OTIF SLA — Supplier receives 180 EDI orders/week from 3 retailers with OTIF SLAs of 96%, 98%, and 99%.

EDI process element	Status before optimization	Improvement implemented
EDI order processing time (receipt to OMS to WMS)	HIGH	4.2 hours average via overnight batch · Urgent orders arrived "the next day"
EDI mapping errors (orders rejected for incorrect format)	MEDIUM	2.8% of orders rejected and requiring manual reprocessing

EDI process element	Status before optimization	Improvement implemented
OTIF risk alerts (orders at risk of arriving late)	HIGH	0% proactive alerts · CS discovered risk only when customer reported the non-compliance
856 management (ASN: dispatch notice to retailer)	MEDIUM	ASN sent with 1.4 hours average delay vs. retailer SLA

Result: OTIF improved from 91.2% to 96.8% in 6 months. Biggest impact: proactive OTIF risk alerts — 68% of at-risk orders were managed before becoming a non-compliance. OTIF penalties: -$280K/year.

WHAT YOU WOULD USE

EDI & B2B OMS

SAP SD + SAP Integration Suite — EDI Integration — Module: EDI Processing + B2B Order ManagementSAP S/4HANA with EDI capabilities for B2B order processing. Manages EDI mappings, automatic order processing, and ASN sending to the retailer.

TrueCommerce / SPS Commerce — Module: EDI Network + Compliance ManagementEDI platforms that connect the supplier with the retailer network. Manage EDI mappings by retailer, ASN generation, and SLA compliance tracking.

HOW IT IS MEASURED

EDI Order Processing Accuracy % (EDI orders processed without mapping error)

EDI Order Processing Accuracy %(EDI orders processed automatically without error or rejection / Total EDI orders received) × 100

Benchmark: >99% accuracy in EDI order processing with well-configured mappings EDI accuracy <97% indicates mapping problems that generate manual work for the CS team — and delays in processing that impact order OTIF.

OTIF Penalty Amount ($) — penalties for B2B customer SLA non-compliance

OTIF Penalty Amount ($)Total penalties billed by B2B customers for OTIF non-compliance in the period

Benchmark: $0 as objective · Any paid penalty indicates OTIF <100% for the corresponding customer Penalty amount is the most direct financial metric of B2B OTIF non-compliance impact. A growing penalty amount requires an urgent root cause analysis.

Automate the B2B order flow and manage to the customer's rules. Map EDI per customer, remove manual entry, alert on SLA risk and turn every chargeback into a fix.

05 Order management analytics: fill rate, OTIF & service metrics

Order management analytics measures how well orders are fulfilled — fill rate, OTIF, backorders, perfect order — by customer, SKU and channel, combining lagging results with leading risk indicators and translating them into revenue terms.

Order management analytics: measuring order fulfillment performance

Order management analytics transform OMS transactional data (orders captured, committed, dispatched, delivered, returned) into actionable insights that improve fill rate, OTIF, and customer experience. Without analytics, the order management team operates reactively — solving problems that already impacted the customer rather than preventing them.

The order management analytics framework

Level 1 — Lagging indicators (what happened): fill rate by SKU and customer, OTIF by channel and carrier, cancelled orders, returns. Level 2 — Leading indicators (what will happen): orders on backorder with projected date, orders at OTIF risk (late picking, delayed in transit). Level 3 — Root cause insights (why it happened): root cause analysis of OTIF non-compliances, fill rate elasticity to inventory level.

How to translate fill rate and OTIF into financial language

94% fill rate with a $12M annual customer = $720K in lost sales from stock-outs. 88% OTIF with a 2% penalty = $240K in penalties on $12M in sales. A 1% fill rate improvement = $120K in incremental revenue. These financial translations connect OMS KPIs to business decisions.

Intermediate vs. Advanced

Intermediate: calculates and reports order management KPIs, presents root cause analysis of key non-compliances.Advanced: designs the order management analytics framework with 3 levels; implements leading indicators for proactive detection; connects OMS KPIs with the CFO's financial language.

BEST PRACTICES

Report fill rate by SKU and customer — the global average hides problems

A 93% average fill rate can hide that 12 SKUs have a 72% fill rate and 3 key customers have 82% OTIF. Fill rate segmentation reveals actionable problems.

Translate fill rate and OTIF into financial language for the executive audience

Fill rate and OTIF are operational metrics. Lost revenue from stock-outs and OTIF penalties are financial metrics. The translation is what connects order management to management's business decisions.

Implement leading indicators (at-risk orders) in addition to lagging (what failed)

Lagging indicators (last month's fill rate) don't prevent the problem — they measure it afterward. Leading indicators (orders on backorder today with tomorrow's delivery date) allow action before the non-compliance occurs.

Automate the OMS dashboard — don't build it manually each month

A manually built Excel dashboard has a 5–10 business-day lag and error risk. A direct OMS connection via Power BI allows automatic, more frequent updates.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Order management dashboard — FMCG company, monthly management review — Monthly OMS dashboard for the Supply Chain Director's management review.

Order management KPI	Current value	Status vs. target
Fill rate % (total, all channels)	MEDIUM	93.4% vs. target 96% — improving from 91.8% previous month
OTIF % (strategic customers with SLA)	HIGH	88.2% vs. target 95% — 3 SLA non-compliances with accumulated penalties: $42K
Backorder rate % (orders on backorder)	MEDIUM	4.2% vs. target <2% — concentrated in 8 SKUs with insufficient safety stock
Perfect order rate % (complete + on-time + no damage + correct documentation)	HIGH	84.6% vs. target 92% — degraded by low OTIF

Result: Narrative: Fill rate improving but OTIF is low from 3 SLA non-compliances generating $42K in penalties. Root cause: 8 SKUs with insufficient safety stock generate 68% of backorders. Action plan: safety stock increase for 8 critical SKUs + review of priority dispatch process for SLA customers.

WHAT YOU WOULD USE

OMS Analytics

Power BI — Order Management Dashboard — Module: Fill Rate + OTIF + Backorder AnalyticsPower BI dashboard connected to the OMS with order management KPIs: fill rate by SKU and customer, OTIF by channel and carrier, backorder aging, and perfect order rate.

SAP Analytics Cloud — Order Fulfillment Analytics — Module: SAP-Integrated Order Management KPIsSAP analytics platform with pre-configured fill rate, OTIF, and backorder reports integrated with SAP S/4HANA and SAP SD.

HOW IT IS MEASURED

Perfect Order Rate % (complete + on-time + no damage + correct documentation)

Perfect Order Rate %(Orders meeting all 4 criteria: completeness + OTIF + no damage + documentation / Total orders for the period) × 100

Benchmark: >92% perfect order rate in B2B distribution · >97% for premium SLA customers Perfect order rate is the most complete order management performance metric. A 97% fill rate with 91% OTIF and 1.5% damage rate results in a perfect order rate <89%.

Revenue at Risk from OTIF/Fill Rate Gap ($)

Revenue at Risk from OTIF/Fill Rate Gap ($)(Customer revenue × (1 - actual fill rate)) + (Customer revenue × (1 - actual OTIF) × % penalty)

Benchmark: $0 revenue at risk from OTIF/fill rate for strategic customers Quantifying revenue at risk converts fill rate and OTIF into financial metrics. “Fill rate is 2 pp below target” is an operational metric. “$240K revenue at risk” is a business metric.

Measure service where the customer feels it. One definition per metric, segmented views, leading indicators and a revenue translation that puts gaps on the leadership agenda.

Pick-Pack-Ship & Fulfillment Execution

The physical execution layer — the processes that convert a committed order into a packed, labeled, and dispatched shipment with the correct product, quantity, and documentation.

01 Picking methods: single, batch, wave & zone picking

Picking methods define how orders are collected: single-order (one order per trip), batch (several orders per trip), wave (orders released in timed groups) and zone (each picker covers one area), often combined.

Picking methods: the choice that determines fulfillment cost and speed

Picking — the activity of retrieving products from the warehouse to prepare orders — represents 50–65% of manual fulfillment operating cost. Selecting the correct picking method for the operation type and order profile can reduce picking cost by 25–50%.

The four main picking methods and their use cases

Single order picking: one operator picks all items for one order at a time — simplest but least efficient (maximum walking distance). Batch picking: one operator picks items from multiple orders in a single warehouse pass — reduces walking distance 40–60% vs. single order. Wave picking: the WMS groups orders into “waves” by destination zone, carrier, or dispatch window — optimizes workflow between picking, packing, and dispatch. Zone picking: each operator works only in their warehouse zone — reduces congestion in large warehouses with many operators.

How to choose the correct picking method

Factors determining the optimal method: (1) Lines per order: few-line orders -> batch picking. Many-line orders -> zone picking. (2) Warehouse size and operator count: large warehouses with many operators -> zone picking to avoid congestion. (3) Speed requirements: same-day delivery -> wave picking with short dispatch windows. (4) Order variability: high variability -> single order or wave. Repetitive orders -> batch.

Intermediate vs. Advanced

Intermediate: executes the assigned picking method at standard productivity, reports process anomalies.Advanced: designs the picking strategy with the correct method by order type and channel; implements wave planning in the WMS; continuously measures and improves LPPH.

BEST PRACTICES

Batch picking is the highest-impact improvement for warehouses with few-line orders (1–5)

In an e-commerce warehouse with 2–3 line average orders, batch picking reduces walking distance 50–65% vs. single order picking with no technology investment — just WMS configuration.

Technology-assisted picking (RF scanner, voice, pick-to-light) improves accuracy more than productivity

Picking technology (RF scanners, voice picking, put-to-light) primarily improves accuracy (eliminating human visual verification errors) and secondarily improves productivity (eliminating paper/list reading time).

Wave planning must be calibrated with carrier dispatch windows

A wave that finishes picking 2 hours before the truck departs generates operators waiting without work. A wave that finishes 30 minutes late generates orders staying in the DC until the next day. Wave planning must calibrate each wave start so picking finishes just in time for the dispatch window.

Zone picking is only efficient when the warehouse has operator density that generates congestion

Zone picking adds coordination complexity (zones must synchronize for the order to be complete) that is not justified if the warehouse doesn’t have a real aisle congestion problem.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> **Illustrative case: Migration from single order picking to batch picking — e-commerce DC, 2,800 orders/day** — DC processes 2,800 orders/day with a mean of 2.3 lines/order using single order picking.

Metric	Single order picking (current)	Batch picking (proposed)
LPPH (lines per hour per operator)	HIGH	48 LPPH — average walking distance: 12.4 km/operator/day
Operators required for daily volume	HIGH	38 picking operators · Labor cost: $84K/month
Optimal batch size (orders per batch)	OK	N/A -> Batch of 8 orders with WMS (item similarity + geographic zone)
Projected LPPH with batch picking	OK	Projected: 78 LPPH (+63%) · Validated in pilot: 74 LPPH (+54%)

Result: Post-implementation: LPPH from 48 to 74 (+54%). Operators required: 38 -> 25 (-13 operators). Monthly labor savings: $28K/month ($336K/year). The 3-week pilot before rollout was critical to calibrate the optimal batch size (8 orders was optimal for this DC's SKU profile).

WHAT YOU WOULD USE

Picking Technology

SAP EWM — Wave Management + Pick by Voice — Module: Wave Planning + Multi-Method PickingSAP EWM supports all picking methods (single, batch, wave, zone) with voice, RF scanner, and pick-to-light assisted picking integration.

Manhattan Associates WMS — Picking Optimization — Module: Intelligent Pick Method Selection + LPPH ManagementManhattan Associates automatically selects the optimal picking method per order based on order profile, operator availability, and dispatch windows.

Pick Assist

Honeywell Vocollect / Zebra — Module: Voice Picking + RF Scanning SolutionsThe reference voice and scanner assisted picking solutions for distribution warehouses.

HOW IT IS MEASURED

Pick Productivity (LPPH — Lines Picked Per Hour)

Pick Productivity (LPPH)Total picking lines completed / Total person-hours dedicated to picking

Benchmark: 60–100 LPPH for single order · 100–180 for batch/wave · 200+ for zone picking in large warehouse An LPPH drop without a volume or order-mix change indicates the picking method or slotting has deteriorated.

Pick Accuracy % (correct picking lines — correct product, correct quantity, correct location)

Pick Accuracy %(Picking lines executed without error / Total picking lines) × 100

Benchmark: >99.9% with scanner picking · >99.5% with voice picking · >98.5% for paper picking (least precise) A picking error rate of 1 in 500 lines seems small, but in a DC with 10,000 lines/day it generates 20 daily errors — each with the cost of correction, reshipping, and customer dissatisfaction.

Let the order profile choose the method. Measure lines per order, match batch, zone or wave accordingly, pilot on real volume and revisit when channels change.

02 Packing station design & cartonization optimization

Packing station design organizes materials and flow so orders are packed quickly and correctly; cartonization selects the right box for each order from item dimensions, fragility and carrier dimensional-weight rules.

Packing: protecting the product, optimizing space, and presenting the brand

The packing station is the fulfillment node where the product collected in picking becomes the finished order ready to dispatch. Packing station design and selection of the correct packaging (carton, polybag, padded mailer, wooden crate) has direct impact on 3 dimensions: product damage rate in transit, packaging cost, and last-mile cost (package size and weight determine carrier rate).

Cartonization: selecting the correct box for each order

Cartonization is the process of automatically selecting the optimal box size for each order's items — minimizing empty space (air) inside the box and maximizing space utilization. The cartonization algorithm: (1) takes the dimensions of each item in the order,

(2) evaluates all available box sizes, and (3) selects the smallest box that can contain all items respecting fragility and stackability constraints.

Cartonization's impact on logistics cost

E-commerce last-mile carrier cost is calculated on the higher of actual weight and dimensional weight (package volume / carrier DIM factor). An oversized box pays for the air space it contains. Correct cartonization can reduce freight cost per order 10–25% by eliminating excess space and selecting the densest possible box.

Intermediate vs. Advanced

Intermediate: operates the packing station per defined standards, selects packaging per process guides.Advanced: designs the packaging strategy with correct cartonization optimization; implements the cartonization algorithm in the WMS; manages the packaging engineering process.

BEST PRACTICES

Cartonization only works if all catalog item dimensions are current

The cartonization algorithm selects the box based on catalog item dimensions. If 15% of items have incorrect or missing dimensions, 15% of orders containing those items will be incorrectly cartonized.

Carrier DIM weight cost is the biggest ROI driver for cartonization investment

E-commerce last-mile carriers charge the higher of actual weight and dimensional weight (volume/DIM factor). An oversized box generates a freight cost that can be 2–3× the freight cost for the product's actual weight.

Design the packing station for flow — not packer convenience

A packing station where the operator must turn to take the box, turn again for fill, and turn again for tape is neither ergonomic nor efficient. U-shaped or linear design aligned with the natural process flow reduces packing time and operator fatigue.

Review the box size catalog when the order mix changes

The optimal box size set changes when the order mix changes (NPIs of different sizes, change in average order lines). An annual catalog review keeps cartonization current.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Cartonization implementation — e-commerce company, 3,400 orders/day — Company uses 12 box sizes but packers select the box manually with their own judgment.

Problem identified	Frequency	Impact
Orders packed in box 2 sizes larger than minimum needed (over-boxing)	38% of orders	Unnecessary DIM weight freight cost: +$0.18/order average
Product damage from too-small box (under-boxing)	4.2% of orders	4.2% damage rate on fragile products -> claims and reshipping
Manual box selection time	42 seconds/order	Packing station de-bottlenecking with cartonization implementation

Result: Post-cartonization implementation in WMS: Over-boxing eliminated in 94% of orders. DIM weight freight cost: -$0.18/order × 3,400 orders/day = $612/day = $220K/year. Damage rate: 4.2% -> 1.1%. Box selection time: 42 sec -> 6 sec (WMS indicates the box before the operator reaches the station).

WHAT YOU WOULD USE

Cartonization

SAP EWM — Packing Station + Cartonization — Module: WMS-Integrated Cartonization AlgorithmSAP EWM manages automatic box selection with integrated cartonization algorithm. The packing operator receives the box instruction before starting the pack.

Packsize / ProShip — Module: On-Demand Packaging + Cartonization OptimizationOn-demand packaging systems (the box is made exactly the size needed for each order) and world-class cartonization platforms.

HOW IT IS MEASURED

Cartonization Accuracy % (orders with the correct box per the algorithm)

Cartonization Accuracy %(Orders packed with the box selected by the cartonization algorithm / Total orders packed) × 100

Benchmark: >95% of orders with the correct algorithm-selected box Cartonization accuracy <90% indicates operators are ignoring the system recommendation — possibly because item dimensions in the product catalog are outdated and the algorithm recommends incorrect boxes.

Packaging Cost per Order (packaging materials cost per order)

Packaging Cost per OrderTotal packaging materials cost (boxes, fill, tape, label) / Total orders packed

Benchmark: $0.80–2.50 per order in FMCG e-commerce · Varies significantly by order density and product type Packaging cost per order in an upward trend without a product type or channel mix change indicates cartonization is selecting unnecessarily large boxes or material prices have risen.

Right box, every time, from good data. Fix master dimensions, rationalize box sizes, let the WMS choose and design stations that make the right action the easy one.

03 Shipping execution: carrier booking, labeling & documentation

Shipping execution covers everything between a packed order and a departed truck: carrier booking, labels and tracking numbers, shipping documents — in Mexico, the CFDI with the Carta Porte complement — and verified loading.

Shipping execution: the last meter of the warehouse before the product leaves

Shipping execution is the set of processes between a packed order and the truck departing from the dispatch dock: carrier selection, shipping label generation, shipment documentation (remittance, bill of lading, CFDI de traslado), and truck loading. In a high-volume DC, shipping execution is the most frequent bottleneck — if it fails, the truck leaves without orders or with incorrect documentation.

Shipping execution steps

(1) Carrier booking: the TMS reserves carrier capacity for the shipment (automatic vs. manual booking). (2) Shipping guide and label generation: the system generates the label with the carrier barcode, delivery address, and tracking number. (3) CFDI de traslado generation (in Mexico, mandatory for merchandise transport in national territory). (4) Remittance or packing list generation for the receiver. (5) Truck loading: verification that all documented packages are loaded.

The CFDI de traslado: Mexico’s fiscal transport requirement

In Mexico, the Comprobante Fiscal Digital (CFDI) de traslado with the Carta Porte complement is mandatory for merchandise transport in national territory (required since 2022, fully enforced since 2024). The CFDI must be issued before transport begins and contains: merchandise description, unit count, value, sender and recipient RFC, transport type, and (in the Carta Porte complement) origin, destination, and transporter’s permit number. Missing the CFDI or Carta Porte complement can result in the SAT stopping the transport.

Intermediate vs. Advanced

Intermediate: executes the shipping process per defined procedure, verifies documentation is correct before authorizing truck departure.Advanced: designs the shipping execution process with correct automation; implements automatic CFDI de traslado generation; manages transport fiscal compliance.

BEST PRACTICES

The CFDI de traslado with Carta Porte must be generated automatically from the ERP — not manually

Manual CFDI de traslado generation is a source of errors (incorrect RFC, incorrect merchandise description, outdated transporter permit number) and delays in shipping execution. Billing system integration with the WMS/TMS generates the CFDI automatically with correct data for each shipment.

WMS-TMS integration for carrier label generation eliminates the biggest shipping bottleneck

When the WMS automatically generates the transport order and the TMS automatically generates the carrier guide and label, shipping execution time is reduced from minutes to seconds per shipment.

Verify truck loading by scanner before closing the dock

Scanner verification that all documented packages are loaded in the truck (and no undocumented extra packages) is the final quality control of shipping execution. Without this verification, “forgotten” packages are only discovered when the customer reports not receiving their order.

The Carta Porte complement requires the transporter’s RFC and valid SICT permit

The Carta Porte complement requires specific transporter data (RFC, SICT permit number) that must be current in the carrier catalog. An expired permit number generates an invalid CFDI.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Shipping execution automation — DC with 840 shipments/day — Current shipping process is 80% manual: guide generated in carrier portal, CFDI in the tax system, packing list printed from ERP.

Process step	Before (manual)	After (automated with TMS + tax integration)
Carrier guide and label generation	12 min/shipment · 2 dedicated FTEs	28 sec/shipment · Automatic generation from integrated WMS/TMS
CFDI de traslado + Carta Porte generation	8 min/shipment · RFC errors: 3.2% of CFDIs	42 sec/shipment · Data from ERP -> 0 RFC errors
Truck loading with scanner verification	Manual count — 18 min per truck of 80 packages	Scanner verification: 6 min per truck + 0 “forgotten” packages
Total shipping execution time per shipment	28 min + errors	5 min + 0 documentation errors

Result: Shipping labor savings: 2 FTEs. Shipping time per shipment: 28 min -> 5 min. SAT transport stops for missing Carta Porte: 0 in the 12 months post-implementation vs. 3 incidents in the prior 12 months.

WHAT YOU WOULD USE

Shipping & CFDI

SAP EWM + SAP S/4HANA — Shipping Integration — Module: Automated Shipping Execution + CFDI de TrasladoSAP EWM + SAP S/4HANA integration automatically generates the remittance, CFDI de traslado with Carta Porte complement, and TMS communication for carrier guide generation.

CONTPAQi / Aspel (Siigo) — Module: CFDI de Traslado + Carta Porte GenerationMost widely used electronic billing systems in Mexico with support for CFDI de traslado generation with the current Carta Porte complement (version 3.1).

Carrier Labels

Shipstation / EasyPost — Module: Multi-Carrier Label Generation APIShipping label generation APIs that connect with multiple carriers and generate labels in the correct format for each carrier.

HOW IT IS MEASURED

Shipping Documentation Accuracy % (shipments with complete and correct documentation)

Shipping Documentation Accuracy %(Shipments dispatched with correct guide + CFDI de traslado + Carta Porte + remittance / Total shipments) × 100

Benchmark: 100% documentation accuracy — any shipment without the correct CFDI is a fiscal risk A shipment without the CFDI de traslado or without the correct Carta Porte complement can be stopped by the SAT in an en-route inspection — with consequences of transport stoppage, fines, and reputational damage.

Shipping Execution Time (time from order ready-to-ship to truck departure)

Shipping Execution TimeAverage time from the last package of a truck being packed to the truck departing from the dispatch dock

Benchmark: <45 minutes for a standard operation · <15 minutes with fully automated shipping execution Slow shipping execution generates trucks departing late from the carrier pickup window — causing delivery delays and potential OTIF non-compliances that are not the carrier’s fault.

Generate documents from data, verify loads by scan. Integrate WMS, TMS and billing, keep carrier data valid and hold documentation accuracy to 100%.

04 Same-day & express fulfillment operations

Same-day and express fulfillment deliver orders within hours of purchase, which requires a cut-off time calculated backward from the delivery window, prioritized processing and inventory close to the customer — urban DCs or stores via ship-from-store.

Same-day fulfillment: the most demanding promise in e-commerce logistics

Same-day delivery (SDD) — delivering the same day the order is placed — is the most demanding fulfillment promise because it requires the entire process (order capture -> picking -> packing -> shipping -> delivery) to occur in hours, not days. The key to SDD is not the speed of each individual step — it's the elimination of wait times between steps.

Critical elements of same-day fulfillment

(1) Cut-off time: the hour until which orders can be captured to guarantee same-day delivery. Cut-off must be calculated backward from the promised delivery window, subtracting last-mile transit time and warehouse fulfillment time. (2) Priority picking: SDD orders must take priority over all others in the WMS queue — with their own wave or explicit prioritization. (3) Correct fulfillment node: SDD is only possible from an inventory node close to the customer (urban DC, physical store doing Ship from Store) — not from a national DC.

Ship from Store: using physical stores as SDD fulfillment nodes

Ship from Store (SFS) is the model where physical stores act as mini-fulfillment warehouses for SDD. The store nearest the customer serves the e-commerce order — drastically reducing last-mile time and cost. SFS requires: (1) real-time store inventory visibility in the OMS, (2) picking process in the store without impacting commercial operations, and (3) a last-mile carrier that picks up at the store.

Intermediate vs. Advanced

Intermediate: manages SDD operation in the assigned DC or store, meets process cut-off times.Advanced: designs the SDD model with correct nodes; implements Ship from Store; manages same-day delivery operation KPIs.

BEST PRACTICES

SDD starts with correct cut-off time design — not with the customer promise

The most frequent SDD mistake: promising a delivery window without verifying whether the required cut-off time is operationally feasible. Calculate real fulfillment time and last-mile time first — then establish the cut-off.

Ship from Store is the most economical way to offer SDD without building an urban DC

An urban DC in CDMX has significant lease and operating cost. Existing physical stores have the space, inventory, and geographic location — they just need the process and technology to act as a fulfillment node.

Measure SDD separately from standard fulfillment

SDD has its own KPIs that must be measured separately: same-day on-time %, same-day fulfillment cycle time, same-day cost per order. Mixing them with standard fulfillment KPIs hides the real SDD performance.

SDD is only sustainable with fulfillment process automation

SDD that depends on operators picking faster is not sustainable. Sustainable SDD requires a fulfillment process designed for speed: slotting of top-selling SKUs in the most accessible zone, exclusive SDD wave planning, and automated shipping execution.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Ship from Store implementation — electronics retailer, 45 stores in CDMX — Retailer wants to offer SDD in CDMX without opening an additional urban DC.

SFS element	Design	Result
SFS-enabled stores (SDD coverage zone)	12 CDMX stores with highest density of nearby e-commerce orders	Coverage: 78% of CDMX postal codes within 45-min delivery from a SFS store
Store inventory visible in OMS in real time	Store WMS -> OMS integration · Update every 15 minutes	Store ATP available for e-commerce channel · 0 overselling since OMS sees store inventory
Picking process in store (without impacting customer service)	Picking windows: 8–11am and 2–5pm · Priority over shelf replenishment	SDD fulfillment time from store: 94 min average vs. 90-min target

Result: SDD from store: 94% of orders delivered before 9pm for orders placed before 4pm. Last-mile cost from store: $28/order vs. $62/order from DC + regular carrier. SDD NPS: 84 (vs. 68 for standard delivery).

WHAT YOU WOULD USE

SDD & Ship from Store

Manhattan Active Omni — Ship from Store — Module: Store Inventory ATP + SDD Order RoutingManhattan Active Omni sees store inventory in real time and automatically routes SDD orders to the nearest store with availability.

Körber / Deposco — Module: SDD Fulfillment WMS for Store OperationsWMS adapted for in-store picking operations within the Ship from Store model.

HOW IT IS MEASURED

Same-Day Delivery On-Time % (SDD orders delivered before the promised window)

Same-Day Delivery On-Time %(SDD orders delivered before [promised time] / Total SDD orders for the day) × 100

Benchmark: >92% of SDD orders delivered within the promised window SDD on-time % <85% indicates the cut-off times, fulfillment times, or last-mile times are incorrectly calibrated and the same-day promise is not sustainable with the current operation.

SDD Fulfillment Cycle Time (time from order capture to delivery)

SDD Fulfillment Cycle TimeTotal time from SDD order entering the OMS to customer delivery confirmation

Benchmark: <4 hours for SDD from urban DC · <3 hours for SDD from Ship from Store SDD fulfillment cycle time measures whether the same-day promise is operationally achievable. A 5-hour cycle time with a 9pm delivery window only allows orders until 4pm.

Engineer the promise before you sell it. Compute the cut-off backward, prioritize same-day work, fulfill from nodes near demand and track same-day performance on its own.

05 Fulfillment KPIs: perfect order, DPMO & order cycle time

Fulfillment KPIs measure the quality and speed of pick-pack-ship: perfect order rate, DPMO (defects per million opportunities), order cycle time, pick accuracy and cost per order, tracked by process stage.

Fulfillment KPIs: measuring end-to-end order execution performance

Fulfillment KPIs measure the quality and efficiency of the entire pick-pack-ship process — from when the WMS receives the picking order to when the order reaches the customer. Without well-defined and systematically measured KPIs, the fulfillment team cannot identify bottlenecks, continuous improvement is impossible, and problems are discovered when the customer reports them.

The fulfillment KPI framework by process

Picking: LPPH, pick accuracy %. Packing: cartonization accuracy %, packaging cost/order, packing time/order. Shipping: shipping documentation accuracy %, shipping execution time, dispatch OTIF. End-to-end quality: perfect order rate %, DPMO (Defects Per Million Opportunities), order cycle time. Productivity: cost per order, cost per line, revenue per FTE.

DPMO: the Six Sigma metric for fulfillment

DPMO (Defects Per Million Opportunities) is the fulfillment quality metric that counts the number of defects per million opportunities — where each picking line is an opportunity and each error (wrong product, wrong quantity, damaged packaging, late delivery) is a defect. A DPMO of 3.4 corresponds to a Six Sigma process (99.99966% accuracy), the target for pharmaceutical or high-precision electronics operations.

Intermediate vs. Advanced

Intermediate: calculates and reports assigned fulfillment KPIs, presents performance analysis in management reviews.Advanced: designs the fulfillment KPI framework with 3 levels; implements automated dashboards; connects fulfillment KPIs with the logistics P&L.

BEST PRACTICES

Perfect order rate is the most useful fulfillment metric for management — not LPPH

LPPH measures internal picking efficiency. Perfect order rate measures whether the customer received what we promised. Only perfect order rate connects fulfillment performance to customer experience.

Calculate DPMO for operations with high quality requirements

For pharmaceutical, precision electronics, or very strict SLA customer operations, DPMO is the correct quality metric. A DPMO of 5,000 in a pharmaceutical operation is unacceptable; in FMCG distribution it may be the starting point.

Order cycle time measures the entire chain's speed — not just the warehouse

A long order cycle time can have its bottleneck in OMS order processing time (before the WMS), carrier transit time (after the WMS), or the picking/packing/shipping process. Stage-by-stage analysis identifies the slowest link.

Automate fulfillment KPIs — don't build them manually

Fulfillment KPIs must be calculated automatically from the WMS, TMS, and OMS. A manually built fulfillment dashboard has a days-long lag and error risk — the opposite of what a continuous improvement process needs.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Monthly fulfillment dashboard — B2B distribution DC, October management review — Cycle summary. All metrics vs. target and vs. previous month.

Fulfillment KPI	Current value	Status vs. target
Pick accuracy % (error-free picking)	MEDIUM	99.2% vs. target >99.5% — improving from 98.8% previous month
LPPH (lines per hour per operator)	MEDIUM	78 LPPH vs. target >85 LPPH — re-slotting opportunity detected
Perfect order rate % (complete + on-time + no damage)	HIGH	88.6% vs. target 94% — limited by carrier OTIF (92.4%)
Order cycle time (capture -> delivery, business hours)	OK	5.8 hours vs. target <6 hours — within target

Result: Narrative: Pick accuracy improving but LPPH still below target — top 40 velocity SKUs detected to have migrated from the golden zone in the last 3 months. Re-slotting scheduled for week 3. Perfect order rate low due to carrier OTIF (92.4% vs. 96% target) — action plan with carrier under review.

WHAT YOU WOULD USE

Fulfillment Analytics

Power BI — Fulfillment KPI Dashboard — Module: Automated Fulfillment ReportingPower BI dashboard with direct WMS, TMS, and OMS connection. Pick accuracy, LPPH, perfect order rate, and order cycle time updated automatically.

SAP EWM Analytics — Module: WMS-Native Fulfillment ReportingNative SAP EWM reports for fulfillment KPIs: per-operator productivity, per-process accuracy, and cycle times by stage.

HOW IT IS MEASURED

Perfect Order Rate % (correct, complete, on-time, no damage, correct documentation)

Perfect Order Rate %(Orders meeting all 5 criteria / Total orders for the period) × 100

Benchmark: >95% perfect order rate in B2B distribution · >99% in pharmaceutical and high-precision electronics Perfect order rate is the most complete end-to-end quality metric. An excellent LPPH with a high damage rate or incorrect documentation results in a low perfect order rate that impacts the customer.

Fulfillment Cost per Order (total fulfillment cost per order)

Fulfillment Cost per Order(Picking + packing + shipping labor cost + materials) / Total orders for the period

Benchmark: $4–12 per order in efficient B2B distribution · $8–25 in e-commerce · Varies by lines/order and automation level Fulfillment cost per order is the financial KPI that connects warehouse operational efficiency with order margin. A $28 fulfillment cost on an order with $22 gross margin destroys value.

Anchor fulfillment on what the customer receives. Perfect order end to end, DPMO where precision is critical, cycle time by stage and dashboards fed automatically from the systems.

Cross-Docking & Flow-Through

The flow-through architecture — eliminating storage where it adds no value, synchronizing inbound and outbound flows, and designing the vendor compliance standards that make it all possible.

01 Cross-docking design: direct, indirect & opportunistic

Cross-docking moves received product straight from the inbound dock to outbound shipping with little or no storage. Three models exist: direct (supplier pre-sorts by destination), indirect (the DC breaks and re-sorts loads) and opportunistic (the WMS diverts a receipt that matches an open order).

Cross-docking: moving product from inbound dock to outbound dock without storing it

Cross-docking eliminates the put-away, storage, and picking steps by moving received product directly to the dispatch dock. It removes the two most expensive warehouse activities (put-away and picking) and drives inventory days at the node toward zero. It is not a universal improvement: cross-docking only works when inbound supply and outbound demand are synchronized and predictable.

The three cross-docking models

Direct (pre-allocated): the supplier ships product already sorted and labeled by final destination. The DC only transfers the pallet from inbound to outbound dock. Lowest cost, highest supplier requirement. Indirect (break-bulk): the DC receives a consolidated load, breaks it down, and re-sorts it by destination. Requires a sorting zone and handling capacity, but works with suppliers that cannot pre-sort. Opportunistic: the WMS detects that a just-received product matches an open order and diverts it to dispatch instead of put-away. Requires no supplier change — only WMS logic.

SKU selection criteria: where cross-docking generates value

Cross-docking candidates share four traits: (1) high and stable rotation — predictable demand that arrives faster than it can be stored; (2) short remaining shelf life, where every stored day destroys value; (3) reliable supplier lead time — variability breaks synchronization; (4) sufficient volume to fill a full outbound movement. SKUs with sporadic demand, unreliable suppliers, or long shelf life are better served by conventional storage — forcing them through cross-dock creates dock congestion without benefit.

Intermediate vs. Advanced

Intermediate: executes cross-dock flows for designated SKUs, escalates when inbound and outbound do not synchronize.Advanced: designs the cross-dock model by SKU and supplier; defines the selection criteria; implements opportunistic cross-dock logic in the WMS; manages the supplier requirements that direct cross-docking depends on.

BEST PRACTICES

Select SKUs for cross-docking on supplier reliability — not just rotation

High rotation makes a SKU attractive; supplier lead-time reliability makes it feasible. A high-rotation SKU from a supplier with variable delivery windows will congest the dock rather than flow through it.

Start with opportunistic cross-docking — it requires nothing from suppliers

Direct cross-docking needs supplier pre-sorting and labeling, which takes months to negotiate. Opportunistic cross-docking is WMS logic alone: when a receipt matches an open order, divert it to dispatch. It is the fastest route to a first result.

Direct cross-docking is a vendor compliance program in disguise

It depends entirely on the supplier shipping pre-sorted, correctly labeled, accurately quantified loads. Without the SSCC and ASN standards behind it, direct cross-docking fails at the receiving dock.

Do not cross-dock a SKU whose demand is not already synchronized with its supply

Cross-docking does not create synchronization — it exploits synchronization that already exists. Applied to unsynchronized flows, it converts a storage problem into a dock congestion problem, which is worse because dock space is scarcer than rack space.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Cross-dock program design — distribution DC, 4,200 active SKUs — The DC stores and picks every SKU. Analysis of rotation, supplier reliability, and shelf life to identify cross-dock candidates.

SKU segment	Model selected	Rationale
High-rotation SKUs from 3 pre-sorting suppliers	Direct cross-dock	Supplier ships pre-labeled by destination · Put-away and picking eliminated entirely
High-rotation SKUs from suppliers without pre-sort capability	Indirect (break-bulk)	Requires sorting zone · Still removes storage and picking
Mid-rotation SKUs with open orders at receipt	Opportunistic	WMS diverts to dispatch when a receipt matches an open order · No supplier change
Low-rotation, long shelf life, variable lead time	Conventional storage	Forcing cross-dock would congest docks without reducing cost

Result (illustrative): Roughly a quarter of volume qualified for some cross-dock model. The decisive constraint was not warehouse capability but supplier lead-time reliability — direct cross-dock was viable only for the three suppliers whose delivery windows were predictable enough to synchronize against.

WHAT YOU WOULD USE

Cross-Dock Execution

SAP EWM — Cross-Docking — Module: Planned + Opportunistic Cross-DockingSAP EWM supports planned cross-docking (pre-allocated at receipt) and opportunistic cross-docking, where the system detects a match between an inbound receipt and an open order and diverts the product to dispatch.

Manhattan Associates WMS — Flow-Through — Module: Cross-Dock Allocation + Dock CoordinationManhattan cross-dock module with inbound-to-outbound allocation logic and coordination against dispatch windows.

HOW IT IS MEASURED

Cross-Dock Rate % (volume moved without put-away)

Cross-Dock Rate %(Units or pallets dispatched without a storage put-away transaction / Total units or pallets received) × 100

Benchmark: Set by the SKU profile, not a universal target · Track the trend against the volume you qualified as cross-dock eligible A cross-dock rate below the eligible volume means qualified SKUs are still being stored — usually because inbound and outbound windows are not synchronized, not because the SKU was wrongly selected.

Cross-Dock Dwell Time (time from inbound receipt to outbound dispatch)

Cross-Dock Dwell TimeAverage time from receipt confirmation to dispatch confirmation for cross-docked units

Benchmark: Hours, not days · Dwell time approaching a full day means the product is being stored, whatever the process is called Dwell time is the honest test of whether cross-docking is real. Product sitting on the dock for 24 hours is storage without racks — it carries the cost of both models and the benefit of neither.

Flow what is synchronized, store what is not. Select by rotation and supplier reliability, start with opportunistic logic, build supplier compliance for direct flows and watch dwell time.

02 Yard management & dock scheduling

Yard management tracks every trailer on site — identity, contents, status and dwell — while dock appointment scheduling controls when carriers arrive and which door they use.

The yard: the blind spot between the road and the warehouse

The yard is where trucks and trailers wait between arriving at the facility and reaching a dock. In most operations it is unmanaged: nobody knows with certainty which trailer holds what, how long it has been waiting, or which dock it should occupy next. The result is trucks queuing while docks stand idle — a coordination failure, not a capacity failure.

The two tools: YMS and appointment scheduling

The Yard Management System (YMS) tracks every trailer on site: identity, contents, status (loaded, empty, awaiting inspection), and dwell time. Appointment scheduling controls arrivals: carriers book a dock and time window in advance rather than arriving whenever they choose. The YMS answers “what is on site right now”; appointment scheduling prevents the site from being overwhelmed in the first place. Neither substitutes for the other.

Why dock scheduling matters more for cross-docking than for storage

A storage warehouse can absorb an early or late truck — the product simply goes to a rack. A cross-dock operation cannot: it depends on inbound and outbound meeting inside a narrow window. Every hour of arrival variability must be absorbed by dock space, labor standing ready, or a broken cross-dock flow. Appointment discipline is therefore a prerequisite of cross-docking, not an optimization on top of it.

Intermediate vs. Advanced

Intermediate: assigns docks and manages the arrival queue, escalates when the yard congests.Advanced: designs the appointment model and dock allocation rules; implements the YMS; manages carrier compliance with booked windows and connects yard performance to the cross-dock and vendor compliance programs.

BEST PRACTICES

Appointment scheduling is worth more than additional dock doors

Most dock congestion is arrival clustering, not insufficient doors. Spreading arrivals across the working day is a scheduling change; adding doors is capital. Exhaust the first before proposing the second.

Put carrier window compliance on the carrier scorecard

A booked window with no consequence for missing it is a suggestion. Window compliance belongs alongside OTIF and damage rate in the D06 carrier scorecard — the same governance, the same volume consequences.

Allocate docks by flow type — not by arrival order

A cross-dock load queuing behind a storage receipt breaks a synchronized flow to serve one that could have waited. Dock priority should follow the cost of delay, and first-come-first-served ignores it entirely.

Measure turnaround by hour of day — the average hides the peak

An acceptable daily mean can contain a three-hour window where turnaround triples. Carriers experience the peak, not your average — and they price accordingly.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Yard and dock scheduling — DC with chronic morning congestion — Carriers arrive unannounced, concentrated in the early morning. Docks sit idle in the afternoon while trucks queue at 7am.

Root cause	Mechanism	Countermeasure
No appointment discipline — carriers arrive at their convenience	Arrivals cluster in a 3-hour morning window	Mandatory booked windows · Unbooked arrivals served last, not first
No trailer visibility — nobody knows what is on site	Loaded trailers wait behind empties · Manual yard checks	YMS with trailer identity, contents and dwell time
Dock assignment is first-come-first-served	Cross-dock loads queue behind low-priority storage receipts	Dock allocation by flow type · Cross-dock loads get window priority
Carrier window compliance is never measured	No consequence for missing a booked slot	Window compliance on the carrier scorecard

Result (illustrative): The congestion was a scheduling problem, not a dock count problem. Spreading arrivals across the working day and prioritizing cross-dock loads relieved the morning queue without adding a single dock door.

WHAT YOU WOULD USE

Yard & Dock

C3 Solutions / Opendock / FourKites Dynamic Yard — Module: YMS + Carrier Appointment SchedulingDedicated yard management and dock appointment platforms with carrier-facing booking portals, trailer tracking and dwell time reporting.

SAP EWM — Yard Management — Module: Yard Management + Door AssignmentSAP EWM yard module with trailer check-in/check-out, yard location management and door assignment integrated with the warehouse process.

HOW IT IS MEASURED

Truck Turnaround Time (gate-in to gate-out)

Truck Turnaround TimeAverage elapsed time from truck gate-in to gate-out

Benchmark: Track by flow type and by hour of day · The distribution matters more than the average — a good mean can hide a 3-hour morning peak Long turnaround has a cost you do not see on your own P&L: carriers price detention risk into their rates. A DC known for slow turnaround pays for it in every freight negotiation.

Appointment Compliance % (arrivals inside the booked window)

Appointment Compliance %(Arrivals within the booked dock window / Total booked arrivals) × 100

Benchmark: High enough that the schedule is trustworthy · Measure by carrier, not only in aggregate Appointment compliance is the input that makes every downstream plan feasible. A cross-dock schedule, a labor plan, and a dispatch wave all assume the truck arrives when it said it would.

Schedule arrivals before adding doors. Mandatory appointments, trailer visibility, door priority by flow and carrier scorecards that count window compliance.

03 Flow-through for retail replenishment

Flow-through is a retail replenishment model in which supplier volume arrives at the DC already allocated to stores, is sorted and shipped the same day, so the distribution node holds almost no inventory.

Flow-through: the DC as a sorting point, not a storage point

In retail flow-through, supplier volume arrives at the DC already destined for specific stores. The DC sorts it and dispatches it the same day. The distribution node holds almost no inventory — it converts inbound supplier loads into outbound store loads. For convenience and supermarket replenishment, where store shelf space is the binding constraint and rotation is high, this is the dominant model.

What flow-through requires that conventional replenishment does not

(1) Store-level demand known before the supplier ships — the allocation happens upstream, not at picking. (2) Suppliers capable of shipping to a schedule reliable enough to sort against. (3) A sorting capability sized for the daily peak, since nothing buffers overnight. (4) Store receiving capable of absorbing daily deliveries. Remove any one and flow-through degrades into cross-dock congestion.

What flow-through moves rather than removes

Flow-through drives inventory days at the distribution node toward zero — but the demand variability that inventory was absorbing does not disappear. It is transferred to the store shelf and to the supplier. This is a good trade when rotation is high and stores replenish daily. It is a poor trade for slow-moving SKUs, where the node was pooling variability across many stores and removing it forces every store to hold its own buffer.

Intermediate vs. Advanced

Intermediate: executes the daily sort and dispatch cycle, manages exceptions when a supplier load is short or late.Advanced: designs the flow-through model by category; sets the store allocation logic; sizes the sorting capability against the peak; and decides which SKUs should stay on conventional replenishment.

BEST PRACTICES

Apply flow-through to rotation, not to the whole catalog

The node earns its cost by pooling variability for slow movers. For fast movers it is mostly handling. Flow-through should remove the handling and leave the pooling in place.

Size the sorting capability against the daily peak

Conventional replenishment lets you catch up tomorrow. Flow-through does not — what is not sorted today misses the store window. Average-based sizing guarantees periodic failure.

Track store on-shelf availability as the real outcome metric

Node DIO reduction is easy to celebrate and easy to achieve by pushing the problem to the shelf. If shelf availability degrades, the inventory was not eliminated — it was relocated to somewhere it works less well.

Flow-through makes supplier reliability a store-level issue

With no node buffer, a late supplier is no longer an inventory problem absorbed by the DC — it is an empty shelf. Supplier lead-time reliability must be qualified before a category moves to flow-through, not diagnosed afterward.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Flow-through for convenience store replenishment — A distribution node serving a high-density convenience network. High-rotation categories with daily store replenishment.

Design decision	Choice	Consequence
Where store allocation is decided	Upstream, before the supplier ships	DC sorts rather than picks · Put-away and picking removed from the flow
Categories on flow-through	High-rotation, daily-replenished	Node DIO near zero · Variability absorbed by shelf and supplier
Categories kept on conventional replenishment	Slow-moving and long-tail	Node keeps pooling variability where the store cannot
Sorting capacity sizing	Sized to daily peak, not average	Nothing buffers overnight — a peak that exceeds sort capacity misses the store window

Result (illustrative): The model works precisely where rotation is high enough that the node was never really buffering — only handling. The failure mode is applying it to the long tail, where the node was doing real risk pooling and removing it pushes buffer stock onto every store at once.

WHAT YOU WOULD USE

Retail Flow-Through

SAP EWM — Cross-Docking + Store Allocation — Module: Flow-Through with Upstream AllocationSAP EWM cross-dock and allocation flows for store-destined volume allocated before receipt, sorted at the node and dispatched the same day.

Blue Yonder / Manhattan Associates — Retail Replenishment — Module: Store Replenishment + Flow-Through AllocationRetail replenishment platforms that compute store-level allocation upstream and drive the sort at the distribution node.

HOW IT IS MEASURED

Flow-Through Rate % (store volume shipped without node storage)

Flow-Through Rate %(Store-destined volume dispatched the same day it was received / Total store-destined volume) × 100

Benchmark: Determined by category rotation and supplier reliability · Measure against the volume you designed for flow-through, not total volume Read this metric alongside store on-shelf availability. A rising flow-through rate with falling shelf availability means the node stopped buffering variability that the store cannot absorb.

Store Delivery Window Compliance %

Store Delivery Window Compliance %(Store deliveries arriving inside the agreed receiving window / Total store deliveries) × 100

Benchmark: High — with no overnight buffer, a missed window becomes a next-day shelf gap In flow-through there is no inventory standing by to cover a miss. Window compliance is not a service nicety; it is the mechanism that keeps the shelf full.

Apply flow-through by rotation, not to the whole catalog. Allocate upstream, size sorting for the peak and judge success by store shelf availability, not node inventory.

04 Merge-in-transit & consolidation hubs

Merge-in-transit consolidates components of one order coming from different origins at a hub, without long-term storage, and ships them together; consolidation hubs more generally combine and redistribute flows between many origins and destinations.

Merge-in-transit: consolidating orders from multiple suppliers in transit

Merge-in-transit (MIT) is the logistics model where components of an order from multiple suppliers are consolidated at a transit hub — without permanent storage — and dispatched together to the final customer. The MIT hub synchronizes multiple origins into the customer's complete order.

Merge-in-transit use cases

MIT is especially valuable when: (1) the customer's order includes items from multiple suppliers that must arrive together. (2) The customer expects a single delivery. (3) Supplier lead times are similar, allowing hub synchronization.

Consolidation hubs: intermediate network nodes for flow aggregation

Consolidation hubs are facilities specifically designed to aggregate flows from multiple origins and redistribute them toward multiple destinations. They are the intermediate nodes in the logistics network that enable: (1) consolidating LTL shipments from multiple suppliers into FTL to destinations, and (2) de-consolidating FTL from plants into LTL for capillary distribution to final customers.

Intermediate vs. Advanced

Intermediate: coordinates supplier arrivals at the consolidation hub, manages exceptions when a component arrives late.Advanced: designs the merge-in-transit model with correct suppliers and flows; implements the hub WMS; manages arrival synchronization and desynchronization risk management.

BEST PRACTICES

MIT requires all supplier lead times to be predictable and comparable

MIT only works when suppliers have sufficiently predictable lead times for the hub to plan synchronization with precision. Suppliers with high lead time variability are incompatible with MIT.

Advance supplier arrival communication (ASN) is critical for MIT

The MIT hub cannot synchronize 3 supplier arrivals without knowing exactly when each one will arrive. The ASN (Advanced Ship Notice) with each supplier's exact ETA is the prerequisite of efficient synchronization.

Define the management protocol when a component arrives late (single point of failure)

If one of the MIT suppliers doesn't arrive in the agreed window, the hub has two options: wait for the component (delaying customer delivery) or dispatch available components (generating an incomplete delivery). The protocol must be defined before go-live.

MIT has the greatest ROI for orders including items from high-volume multi-supplier

MIT doesn't justify its cost for sporadic or low-volume orders. MIT ROI scales with the volume of multi-supplier orders that can be consolidated at the hub.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Merge-in-transit for white goods dispatch — electronics distributor — A white goods distributor receives orders for complete sets (refrigerator + washer + dryer) from 3 different suppliers.

MIT component	Design	Result
MIT hub in Querétaro (equidistant between 3 suppliers)	4,200 m² hub · No storage racks — only transit zone and docks	Hub lease cost: 40% lower than a standard DC per useful m²

MIT component	Design	Result
Synchronization of 3 supplier arrivals	Coordinated arrival windows: 8–10am · Complete set consolidated before 11am	98.2% of sets consolidated in <3 hours of last component arrival
Complete set dispatch to customer on a single truck	FTL from hub to final customer · Delivery within 24 hours from hub	Customer satisfaction: complete set delivered in one event vs. 3 separate deliveries before MIT

Result: Delivery customer NPS: from 62 (3 separate deliveries) to 84 (complete set delivery). Logistics cost per set: -12% from FTL consolidation vs. 3 supplier LTLs to distributor DCs.

WHAT YOU WOULD USE

Merge in Transit

SAP EWM — Merge-in-Transit Processing — Module: Multi-Supplier Consolidation + WMS IntegrationSAP EWM supports MIT processing: consolidation hub management, per-order component tracking, and complete set dispatch.

HOW IT IS MEASURED

MIT Assembly Accuracy % (MIT orders completed with all correct components)

MIT Assembly Accuracy %(MIT orders dispatched with all order components correctly consolidated / Total MIT orders) × 100

Benchmark: >99% accuracy in merge-in-transit · A MIT error (missing a component) generates a complete order delivery failure A MIT error resulting in incomplete set delivery to the customer is significantly more costly than a standard picking error — because the customer is already expecting the complete set and an incomplete delivery requires a second delivery event.

MIT Dwell Time (time from first component arrival to complete set dispatch)

MIT Dwell TimeAverage time from first set component arriving to the complete set being dispatched

Benchmark: <4 hours for well-synchronized MIT · >8 hours indicates supplier synchronization problems MIT dwell time measures the effectiveness of supplier synchronization at the hub. A long dwell time means one or more suppliers are arriving significantly before others — generating wait from first to last component.

Merge only what you can synchronize. Qualify suppliers on reliability, run on accurate ASNs, decide the late-component rule upfront and measure dwell time relentlessly.

05 Vendor compliance programs: SSCC, ASN & store-ready merchandise

Vendor compliance programs define the standards suppliers must meet so receipts can be processed efficiently — GS1 SSCC pallet labels, accurate advance ship notices, correct quantities, delivery windows and, in retail, store-ready merchandise — with measurement and chargebacks.

Vendor compliance: the standards the supplier must meet for the DC to run well

Vendor compliance is the set of standards and requirements that the distributor or retailer imposes on its suppliers to ensure product arrives in the correct conditions to be processed efficiently in the DC — correct labeling, complete documentation, adequate packaging, and exact quantities. Supplier non-compliance generates significant hidden costs: warehouse rework, receiving delays, and inventory errors.

The most important vendor compliance standards

(1) SSCC (Serial Shipping Container Code): the GS1 barcode on the pallet that identifies the exact content (supplier, item, lot, quantity). The SSCC enables automatic pallet scanning without opening the box. (2) ASN (Advanced Ship Notice): the electronic shipment notification the supplier sends before the truck arrives at the DC — with exact detail of what's coming. (3) Store-ready merchandise: product already labeled with the price and destination store code, ready to go directly to the shelf without rework.

The cost of supplier non-compliance

Supplier non-compliance has direct and indirect costs: (1) Warehouse rework (re-labeling packages, separating mixed pallets). (2) Receiving delays (which generate delays in inventory availability). (3) Inventory errors (incorrect SSCC generates discrepancies between system and physical inventory). Large retailers commonly charge suppliers for non-compliance (labels, ASN, delivery windows); the rules and amounts are set in the supplier manual of each retailer.

Intermediate vs. Advanced

Intermediate: verifies compliance of assigned suppliers, reports non-compliances to the procurement team.Advanced: designs the vendor compliance program with correct standards and penalties; implements the chargeback process; manages the compliance improvement program with suppliers.

BEST PRACTICES

Implement chargebacks for non-compliance — without consequences, compliance doesn't improve

Supplier non-compliance is rational if it has no consequences: correctly labeling pallets has a cost for the supplier. Chargebacks transfer that cost to the supplier when they don't meet standards.

The vendor compliance program must start with communication and training — not just penalties

Suppliers that don't meet standards often don't know them well or don't have the systems to meet them. The first step is communicating standards clearly and allowing adaptation time — penalties come after.

Integrate vendor compliance KPIs into the supplier scorecard of the procurement team

Vendor compliance is part of supplier performance — as important as OTIF and product quality. A supplier with 100% OTIF but 35% non-compliance is generating significant hidden costs in the warehouse.

Report non-compliance by supplier and type — aggregation hides problems

A 12% average non-compliance rate can hide that 3 suppliers have 85% of incidents. Segmentation by supplier and non-compliance type identifies actionable problems.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Vendor compliance program — distributor with 180 active suppliers — Analysis of supplier non-compliance impact on DC operations.

Non-compliance type	Frequency (% of receipts)	Rework cost
Incorrect or missing box label (no SSCC or unreadable barcode)	22% of receipts	$0.85/box in labeling rework · $42K/month in the DC
ASN not sent or sent with incorrect data	38% of suppliers without ASN	4.2 additional hours of manual receiving per truck vs. 1.4 hours with ASN
Incorrect quantity on pallet (difference between document and physical)	8% of pallets	18 minutes of recount and adjustment per affected pallet · Inventory discrepancies

Result: Vendor compliance program implementation with chargebacks: $0.50/box for incorrect label, $120/truck without ASN, $25/pallet with incorrect quantity. In 6 months: label non-compliance: 22% -> 6%. ASN compliance: 62% -> 91%. DC rework: -$38K/month.

WHAT YOU WOULD USE

Vendor Compliance

SAP SD — SSCC + ASN Management — Module: EDI + SSCC Barcode + ASN ProcessingSAP S/4HANA manages the vendor compliance process: ASN receipt via EDI, SSCC verification when scanning the pallet, and automatic chargeback generation for non-compliance.

TrueCommerce / SPS Commerce — Module: EDI + Vendor Compliance PortalEDI platforms with vendor compliance portal. Suppliers can view their compliance standards, ASN status, and pending chargebacks.

HOW IT IS MEASURED

Vendor ASN Compliance % (suppliers sending correct and timely ASN)

Vendor ASN Compliance %(Receipts with correct ASN sent before truck arrival / Total receipts for the period) × 100

Benchmark: >95% of suppliers sending correct, timely ASNs Without ASN, receiving is manual — the warehouse team opens each box to count the content. With high-quality ASN, scanning the pallet's SSCC automatically confirms the complete receipt.

Non-Compliance Chargeback Amount ($) per supplier

Non-Compliance Chargeback Amount ($) per supplierTotal chargebacks generated to the supplier for vendor compliance standard non-compliance

Benchmark: $0 as objective for each supplier · Suppliers with recurring chargebacks must have a formal improvement plan Chargebacks are the financial consequence of non-compliance that creates the right incentive for improvement. Without chargebacks, non-compliance has no consequences and DC rework continues indefinitely.

Set the standard, help suppliers meet it, then enforce it. Publish the manual, train, measure every receipt and use chargebacks to keep non-compliance from becoming your cost.

Warehouse Automation & AMR/Robotics

The technology layer that is transforming warehouse operations from labor-intensive to capital-intensive — with the business case framework that determines when, what, and how much to automate.

01 AMR & mobile robotics: autonomous mobile robots in the warehouse

Autonomous mobile robots (AMRs) navigate the warehouse with sensors and maps, without fixed guides, to assist picking — bringing totes or shelves to people, following pickers, or moving loads between zones.

AMRs: the robots changing the collaborative picking paradigm

AMRs (Autonomous Mobile Robots) are robots that navigate autonomously in the warehouse to assist operators in picking — bringing the collection container to the operator, guiding them to the next picking location, or transporting boxes between stations. Unlike AGVs that follow fixed routes, AMRs navigate autonomously using sensors and warehouse maps, adapting in real time to obstacles.

AMR applications in the warehouse

Goods-to-operator (GTO): the robot brings the shelf or container to the operator — the operator doesn't walk, the robot does. Productivity: 2–4× the LPPH of manual picking. Follow-me: the robot follows the operator carrying the collection container. Transportation AMR: transports pallets or containers between warehouse zones — replaces manual pallet jacks. Inventory scanning AMR: autonomously travels the warehouse scanning barcodes at each location to verify inventory.

The AMR business case

AMRs have an acquisition cost of $20,000–80,000 per robot with an annual maintenance contract of 8–12% of cost. ROI comes from: (1) increased LPPH of operators working with AMRs (2–4×), (2) elimination of walking-related accidents and fatigue, and (3) flexibility to scale capacity by renting or buying more AMRs without infrastructure works.

Intermediate vs. Advanced

Intermediate: operates available AMRs, reports productivity KPIs of operators working with AMRs.Advanced: designs the AMR business case; leads the supplier selection process; manages the implementation project and WMS integration.

BEST PRACTICES

Collaborative AMRs have the fastest ROI of all automation technologies

Collaborative AMRs don't require warehouse redesign or the infrastructure of complete GTP systems. They are implemented in weeks, not months.

WMS-AMR integration is critical for optimal robot routing

The WMS must communicate to the Fleet Management Software which picking missions are available so the software assigns the correct AMR to each mission based on its position and mission priority.

AMR and human operator coexistence requires clear safety rules

AMRs must operate in zones where operators know the coexistence rules (maximum speeds, AMR zone signage, emergency protocols).

Scale the AMR fleet with business growth — flexibility is the greatest advantage

AMRs can be added or redistributed between zones in hours — the greatest advantage vs. fixed systems in environments with seasonal demand or rapid growth.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> **Illustrative case: Collaborative AMR implementation — e-commerce DC, 8,200 lines/day** — 18 Goods-to-Operator AMRs for picking of the 2,400 highest-velocity SKUs.

Metric	Before (manual picking)	After (AMR-assisted picking)
Picking operator LPPH	OK	64 LPPH -> 168 LPPH (+163%)
Operator walking distance/day	OK	14.2 km/day -> 2.8 km/day (-80%)
Picking errors (incorrect lines)	OK	1.4% error rate -> 0.3% error rate (-79%)
Operators required for volume	OK	42 operators -> 18 operators + 18 AMRs

Result: ROI at 24 months: $1.8M in labor savings + $420K in picking error reduction vs. $1.4M investment in 18 AMRs. Payback: 14 months. Picking error reduction was the least expected but most valuable benefit.

WHAT YOU WOULD USE

AMR Providers

Locus Robotics / other collaborative AMR vendors — Module: Collaborative AMR for Order PickingAmong the most widely used collaborative AMR providers in North American e-commerce.

Geek+ / HAI Robotics — Module: Goods-to-Person AMR + ASRSChinese-origin AMR providers with high-density GTP systems.

HOW IT IS MEASURED

AMR-Assisted LPPH (productivity of operators working with AMRs)

AMR-Assisted LPPHTotal lines picked by operators with AMR / Total hours of AMR-assisted operators

Benchmark: 150–250 LPPH for Goods-to-Operator AMRs · 2–4× manual picking LPPH AMR-Assisted LPPH <120 for a GTO system indicates configuration problems, slotting quality issues, or operator learning curve problems.

AMR Fleet Utilization %

AMR Fleet Utilization %(AMR hours on active mission / Total available hours of the AMR fleet) × 100

Benchmark: >70% fleet utilization in mature operations Low utilization indicates more robots than needed for current volume — or the mission assignment process is not optimized.

Use AMRs to remove walking without rebuilding the building. Pick the right use case, prove productivity in a pilot, integrate with the WMS and grow the fleet with demand.

02 Automated Storage & Retrieval Systems (ASRS)

Automated Storage and Retrieval Systems (AS/RS) store and retrieve pallets, totes or cases with cranes, shuttles or robotic grids under software control, using building height to achieve very high storage density.

ASRS: high-density automated storage

ASRS systems automatically store and retrieve pallets or boxes using cranes or shuttles controlled by software — without human intervention. ASRS are justified when the warehouse m² cost is high and when the volume of movements is sufficient to utilize the system's capacity.

Types of ASRS

Pallet ASRS: a crane retrieves the pallet from the correct cell. Can store pallets up to 40 meters high, tripling m² storage capacity. Mini-load ASRS for boxes: high-speed shuttles for boxes — the highest-density solution for e-commerce or pharmaceuticals. AutoStore: 3D grid container storage — very high density (several times conventional shelving).

The ASRS business case: when density justifies the investment

ASRS is justified when: (1) warehouse m² cost is >$400/year (urban zones), (2) movement volume is >500 pallets/day, and (3) the operation requires high retrieval speed and high inventory accuracy. It's the highest capex investment in automation but has the lowest operating cost per movement once installed.

Intermediate vs. Advanced

Intermediate: operates the available ASRS per defined procedures, reports system anomalies.Advanced: designs the ASRS business case; leads supplier selection and layout design; manages the implementation project.

BEST PRACTICES

ASRS has the greatest ROI in warehouses where m² is the scarcest resource

In urban zones where m² cost is very high, ASRS can generate a 3–5 year ROI even with high capex.

Always design a manual operation alternative for when the ASRS fails

A warehouse that depends 100% on the ASRS and has no manual operation alternative has a total shutdown risk. The design must include minimum space for an emergency manual operation.

ASRS-WMS integration requires specialized middleware (WCS)

The WCS (Warehouse Control System) manages crane movement. Correct WMS-WCS integration ensures ASRS movements are coordinated with WMS picking orders.

The ASRS maintenance contract is as important as the system price

Preventive and corrective maintenance with availability and response time SLAs is the factor that determines the system's real useful life.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Pallet ASRS — pharmaceutical warehouse in urban zone, limited space — The warehouse needs to triple its capacity without expanding m².

Alternative	Capacity in 3,200 m²	Estimated capex
Additional conventional shelving	MEDIUM	2,100 pallets (vs. 1,800 current) · Only +17% capacity
Pallet ASRS at 28 meters height	OK	6,400 pallets (+256%) · No footprint expansion
vs. Additional 4,500 m² lease	HIGH	Equivalent · $7.8M in 5 years of urban zone lease

Result: Selection: Pallet ASRS at 28 meters. Capex: $4.2M. ROI vs. additional lease: $7.8M - $4.2M = $3.6M in 5 years. Payback: 3.2 years. 6,400-pallet capacity in the existing footprint was the decisive argument.

WHAT YOU WOULD USE

ASRS Providers

Dematic / SSI Schaefer — Module: Pallet ASRS + Mini-Load ASRSTwo of the largest warehouse automation providers globally.

AutoStore — Module: Grid-Based High-Density StorageVery high-density grid storage system. Especially suitable for pharmaceutical e-commerce and high-precision electronics.

HOW IT IS MEASURED

ASRS Throughput (movements per hour)

ASRS ThroughputTotal storage and retrieval movements / ASRS operating hours

Benchmark: Pallet ASRS: 60–120 movements/hr · Mini-load: 200–500 movements/hr · AutoStore: 300–800 movements/hr Throughput below manufacturer specs may indicate software problems, mechanical failures, or a demand profile incompatible with the system design.

ASRS Availability % (% of time the ASRS is operational without failures)

ASRS Availability %(ASRS operating hours without failure / Total planned operating hours) × 100

Benchmark: >98% availability for ASRS in critical operations An ASRS with <95% availability in a pharmaceutical or high-speed e-commerce warehouse is a serious operational risk — the warehouse can be paralyzed if the ASRS fails without a manual alternative.

Build up only when the numbers and the volume are durable. Compare against alternatives, size for the peak, keep a manual fallback and contract availability, not just equipment.

03 Pick-to-light, put-to-light & voice-directed picking

Assisted picking technologies guide the operator to the right item and quantity: pick-to-light and put-to-light use lights at locations, voice picking gives spoken instructions, and RF scanners verify each pick by barcode.

Assisted picking technologies: guiding the operator to eliminate errors

Assisted picking technologies guide the operator with visual or voice signals to execute the correct pick without needing to read a paper list or screen. The guidance eliminates human picking errors while increasing speed.

The three assisted picking technologies

Pick-to-light: a light at the correct location tells the operator what to take and how much — the operator just follows the lights. Pick accuracy: >99.9%. Ideal for high-speed warehouses with few locations. Put-to-light: the opposite of pick-to-light — the operator brings product to the correct location guided by the light. Ideal for returns sorting or store put-away. Voice-directed picking: voice picking instructions to the operator (hands-free, eyes-free). Pick accuracy: >99.9%. The industry standard for high-volume picking.

When to justify each technology

Pick-to-light: critical accuracy and fixed, numerous locations (pharmaceutical, electronics). Voice-directed picking: the highest ROI assisted picking technology for mid-to-high volume — more flexible than pick-to-light. RF scanner: the lowest-cost assisted picking technology, the baseline for all modern warehouses.

Intermediate vs. Advanced

Intermediate: operates available assisted picking technologies, reports accuracy and productivity KPIs.Advanced: designs the assisted picking strategy with the correct technology by zone; evaluates ROI; leads implementation.

BEST PRACTICES

Voice-directed picking is the highest ROI assisted picking technology for most warehouses

More flexible than pick-to-light, with similar accuracy (>99.9%), and improves productivity 15–30% vs. RF scanner.

Pick-to-light is the correct technology for high-speed zones with fixed SKUs

In zones where the same SKUs are picked with very high frequency and locations are fixed, pick-to-light is faster than voice-directed.

Integrate assisted picking technology with the WMS for automatic confirmation

Pick confirmation must be recorded automatically in the WMS — without an additional operator action.

Measure pick accuracy by technology, zone, and operator to identify actionable problems

DC average accuracy can hide that a specific zone or operator has significantly lower accuracy.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Voice-Directed Picking — pharmaceutical distribution DC, 12,000 lines/day — Migration from RF scanner to voice-directed picking.

Metric	RF Scanner (before)	Voice-Directed Picking (after)
Pick accuracy %	MEDIUM	99.2% — 58 errors/day at 12,000 lines
Pick accuracy % with voice	OK	99.97% — 4 errors/day (-93% errors)
LPPH (lines per hour)	OK	68 LPPH -> 84 LPPH (+24%)

Result: Voice-directed picking ROI at 18 months: $480K in error correction cost reduction + $220K in additional productivity vs. $180K voice system investment.

WHAT YOU WOULD USE

Pick Assist Technology

Zebra Technologies / Honeywell Vocollect — Module: RF Scanners + Voice-Directed Picking HardwareThe reference assisted picking hardware providers. Honeywell Vocollect leads in voice-directed; Zebra leads in RF scanners.

Knapp / Dematic — Module: Pick-to-Light + Put-to-Light SystemsThe providers of pick-to-light and put-to-light systems for high-speed warehouses.

HOW IT IS MEASURED

Directed Pick Accuracy %

Directed Pick Accuracy %(Lines executed correctly with the technology / Total technology-assisted lines) × 100

Benchmark: >99.9% for pick-to-light and voice-directed · >99.5% for RF scanner Directed pick accuracy <99.5% with voice-directed indicates operators are confirming locations without physically verifying the product — a process problem the system alone cannot solve.

Voice-Directed LPPH vs. RF Scanner LPPH

Voice-Directed LPPH vs. RF Scanner LPPHLPPH with voice-directed / LPPH with RF scanner

Benchmark: 15–30% LPPH improvement migrating from RF scanner to voice-directed LPPH improvement comes primarily from eliminating screen reading time and manual confirmation. The voice operator has hands and eyes free for picking.

Guide the picker, then verify the pick. Baseline errors, choose lights, voice or RF by zone, integrate with the WMS and audit where accuracy slips.

04 Conveyor systems, sorters & warehouse control systems (WCS)

Conveyors move cases and totes between zones, sorters divert them to destinations at high speed, and the Warehouse Control System (WCS) coordinates this equipment in real time, translating WMS instructions into machine commands.

Automatic transport systems: flow without human intervention

Conveyor systems (conveyor belts) and sorters (automatic classifiers) are the infrastructure that moves boxes and packages between zones without human intervention. In high-volume warehouses, conveyors and sorters are the infrastructure that makes the fulfillment speed achievable that manual systems cannot reach.

Types of conveyor and sorter systems

Accumulation conveyor: allows flow control and zone synchronization. Crossbelt or tilt-tray sorter: can process 5,000–10,000 items per hour with 99.99% accuracy. Scan-weigh-measure (DWS): measures the volume, weight, and barcode of each package on the conveyor line — required for carrier billing and content verification.

The WCS: the software that controls the automation infrastructure

The Warehouse Control System (WCS) manages conveyor and sorter systems in real time — controlling speeds, each box's route, and fault diagnostics. It sits between the WMS (which manages inventory and orders) and the hardware (conveyors, sorters, AMRs) — translating WMS business instructions into control commands for the hardware.

Intermediate vs. Advanced

Intermediate: operates the available conveyor and sorter system, reports incidents and system performance metrics.Advanced: designs the conveyor/sorter system layout; manages the WCS maintenance contract; leads infrastructure expansion or modification projects.

BEST PRACTICES

Sorters only have ROI with sufficiently high volumes — don't underestimate requirements

A mid-capacity crossbelt sorter costs $2–5M. ROI is only achieved with >20,000 items/day and many classification destinations.

Preventive sorter maintenance is the most important investment for availability

A sorter that stops at peak volume has the highest possible operational impact.

The WCS must be integrated with the WMS for correct routing of each box

Without WMS-WCS integration, the sorter doesn't know where to send each box because it doesn't know the box content.

Monitor the sorter reject bucket — it's the indicator of data quality problems

Boxes going to the reject bucket are those the system couldn't identify. A high reject bucket indicates a labeling or master data problem.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Crossbelt sorter — FMCG distribution DC, 35,000 boxes/day — Current sorting process is manual with 80 operators.

Metric	Manual sorting	Crossbelt sorter
Sorting capacity (boxes/hour)	HIGH	3,200 boxes/hour with 80 operators
Sorter capacity	OK	9,600 boxes/hour with 3 operators (1 supervision + 2 feeding)
Sorting accuracy (box to correct destination)	OK	97.8% manual (1,540 errors/day) -> 99.99% (0.35 errors/day)

Result: Sorter implemented: Capacity +200%. Sorting operators: 80 -> 3. ROI at 3 years: $3.6M labor savings + $1.2M error reduction vs. $4.2M investment. Payback: 2.6 years.

WHAT YOU WOULD USE

Conveyor & Sorter

Interroll / Vanderlande — Module: Conveyor Systems + Crossbelt SortersThe global reference conveyor and sorter system providers. Vanderlande is part of Toyota Industries.

Dematic / SSI Schaefer — Module: WCS + Integrated AutomationProviders offering both automation hardware and integrated WCS.

HOW IT IS MEASURED

Sorter Throughput (boxes sorted per hour)

Sorter ThroughputTotal boxes processed by the sorter / Operating hours

Benchmark: 5,000–15,000 boxes/hour depending on type · Real throughput must be >80% of manufacturer nominal Real throughput <70% of nominal indicates feeding problems, code reading problems, or mechanical issues.

Sorter Accuracy %

Sorter Accuracy %(Boxes sent to correct destination / Total boxes processed by sorter) × 100

Benchmark: >99.9% accuracy for correctly calibrated sorters Sorter accuracy <99.5% indicates barcode reading problems that generate boxes to the reject bucket or incorrect destination.

Automate flow where volume is sustained, and protect it. Justify on volume, integrate WMS and WCS carefully, fund maintenance and watch the reject lane as a data-quality gauge.

05 Automation ROI, total cost of ownership & implementation roadmap

The automation business case compares the total cost of ownership of automated options with continuing manual operation over the same horizon, and sequences investments in a roadmap from lower to higher capital and risk.

Warehouse automation ROI: the framework for deciding what, when, and how much to automate

Warehouse automation is not a binary decision — it's a continuous decision about which processes to automate, with which technology, and in which sequence. The correct framework evaluates the ROI of each option considering full TCO, implementation time, and operational risk.

Automation TCO: beyond the initial capex

Full TCO includes: (1) Capex: acquisition and installation cost. (2) WMS/ERP integration: can be 20–40% of capex. (3) Annual maintenance: 8–12% of capex, year after year. (4) WCS and robot software updates. (5) Implementation and training costs. (6) Unplanned downtime cost. An analysis that only considers capex underestimates real TCO by 40–60%.

The automation decision framework: 5 criteria

(1) Volume: is it high enough to utilize the system's capacity? (2) Demand variability: can the technology handle peaks? (3) Labor cost: is it high enough to justify the capex? (4) m² cost: is space the scarce resource? (5) Payback timeframe: does the CFO accept projects with payback >3 years?

Intermediate vs. Advanced

Intermediate: supports the evaluation process with current operation productivity and cost data.Advanced: designs the automation business case with correct TCO; leads the supplier selection process; manages the phased implementation roadmap.

BEST PRACTICES

Evaluate the full TCO — not just automation capex

Annual maintenance at 10% of capex, accumulated over 5 years, is 50% of the original capex. Omitting it underestimates the real TCO and generates incorrect investment decisions.

Implement automation in phases — from lowest to highest cost and risk

Starting with the lowest-capex, highest-ROI technologies (voice-directed, AMRs) before higher-capex ones (ASRS, sorters) allows learning with real data before committing major capital.

Use real operation productivity and cost data in the business case

A business case based on industry benchmarks instead of real data can have ±30–50% errors in projected savings.

The automation business case must include the "do nothing" scenario

The base scenario must compare automation TCO vs. manual operation TCO over the same period — including projected labor cost growth.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Automation roadmap — distribution DC, 3 phases over 5 years — Evaluation of a 3-phase, 5-year automation investment roadmap.

Phase and technology	Capex	Projected ROI
Phase 1 (Year 1): Voice-directed picking + RF scanners in all zones	$180K	Pick accuracy: +0.7 pp · LPPH: +24% · Payback: 14 months
Phase 2 (Year 2): 12 collaborative AMRs for top 40% of picking volume	$840K	LPPH: +80% in AMR zones · -18 operators · Payback: 22 months
Phase 3 (Year 4–5): Pallet ASRS for bulk zone (if volume justifies)	$3.2M	Bulk DC DIO: -30% · Cost per pallet: -42% · Payback: 36 months

Result: Roadmap approved. Phase 1 generates the fastest ROI (payback 14 months) with the least risk. Phase 3 is conditioned on Year 3 volume exceeding 1,200 pallets/day — the threshold that justifies the ASRS.

WHAT YOU WOULD USE

Automation Consulting

Gartner Magic Quadrant for Warehouse Management Systems — Module: Automation Market IntelligenceGartner publishes a Magic Quadrant for Warehouse Management Systems and market guides on warehouse automation — useful as a starting point for a vendor long list, not as a decision.

Interact Analysis / LogisticsIQ — Module: Warehouse Automation AnalyticsSpecialist analytics firms in the automation market with ROI benchmarks by technology.

HOW IT IS MEASURED

Automation Investment ROI %

Automation Investment ROI %(Total savings generated by automation / Total cost [capex + opex]) × 100

Benchmark: >200% ROI in 5 years to justify a warehouse automation project An automation project with <150% ROI in 5 years is high risk — payback is >3.3 years and assumes labor cost and volume will remain stable throughout that period.

Automation Payback Period (months to capex recovery)

Automation Payback Period(Automation capex / Monthly savings generated by automation)

Benchmark: <24 months for low-risk projects (AMRs, voice-directed) · <36 months for high-investment projects (ASRS, sorters) Payback period is the most commonly used indicator by CFOs to approve automation projects.

Decide automation on full TCO and phase it. Use your own data, count every cost, compare with the manual future and tie later phases to volume gates.

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